Behavioural Responses to Housing Subsidies: Evidence from Ireland's Help to Buy Scheme

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Abstract

When a housing subsidy is targeted, restricted to a particular buyer group, dwelling type, or price range, it can distort behaviour in two distinct ways. At the eligibility boundary, sellers have an incentive to keep transactions just below the cutoff. Within the eligible segment, the composition of buyers shifts towards those the subsidy was designed to help. This paper studies both responses in the context of Ireland's Help to Buy scheme, a tax-based subsidy for first-time buyers of new-build dwellings valued at or below €500,000. Using a developer-side notch model alongside evidence from the Property Price Register and administrative records of household purchases, I find clear bunching just below the €500,000 cutoff in the new-build market under both policy regimes, with no such pattern in second-hand transactions or before the policy was implemented. The marginal developer's implied transaction-value response is around one-third of the maximum statutory relief in each regime. Over the same period, the first-time buyer share of new-build purchases grows relative to the second-hand market, and the new-build share of first-time buyer purchases grows relative to former owner-occupiers, with both responses roughly doubling when the marginal subsidy rate rises from 5% to 10%. The threshold distortion and the participation response strengthen together as the subsidy becomes more generous. Targeting a housing subsidy therefore involves an unavoidable trade-off: the same design choices that direct support towards intended buyers also concentrate behavioural distortion at the eligibility boundary, and both effects intensify with the size of the subsidy.

1 Introduction

Housing purchase subsidies are a common policy instrument for promoting home-ownership and supporting housing market activity, particularly amongst younger and liquidity-constrained households. Yet when eligibility depends on a transaction-value cutoff, the policy creates a discontinuous incentive to remain below that threshold, and when it is also restricted to a particular buyer group or dwelling type, it alters who participates in the treated segment. A growing empirical literature documents threshold responses in housing markets (Agarwal, Hu, and Lee, 2023; Uribe, 2022; Best and Kleven, 2018; Kopczuk and Munroe, 2015), and Hargaden (2017) shows that buyer-specific tax exemptions can generate allocative distortions across buyer types at a threshold. Less is known, however, about how these two margins interact when a subsidy is targeted on a subset of demand within a defined market segment: whether the local threshold distortion is attenuated by the restricted coverage, whether the participation margin responds at all, and whether the two respond together as subsidy generosity rises.

This paper studies Ireland's Help to Buy (HTB) scheme, a tax-based subsidy for first-time buyers of new-build dwellings with a transaction-value at or below €500,000. Introduced in January 2017, the scheme initially provided relief equal to 5% of the transaction value, capped at €20,000, and was enhanced in July 2020 to 10%, capped at €30,000. Second-hand properties have never been eligible. This setting is well suited to studying the two margins jointly: the policy creates a salient notch at €500,000 in the new-build market, leaves a natural ineligible comparison group in the second-hand market, and contains a discrete increase in subsidy generosity within the same institutional framework. Unlike much of the existing notch literature, which studies transaction taxes or price ceilings that apply to the full market, HTB applies only to a subset of buyers, so the size of the threshold distortion depends on the share of demand that is sensitive to losing eligibility at the cutoff.

To study these issues, I propose a developer-side notch model that makes the link between the two margins explicit, and combine it with three empirical strategies. First, using transaction-level data from the Property Price Register, I estimate bunching at the €500,000 threshold with the polynomial counterfactual method of Best and Kleven (2018). Second, I implement a distributional difference-in-differences comparing new-build and second-hand transactions across the two reforms, locating the threshold response within the wider distribution and distinguishing a sharply local distortion from diffuse adjustment under optimisation frictions. Third, using administrative records of household purchases from the Central Statistics Office, I estimate two buyer-participation difference-in-differences designs to test whether HTB increased the relative importance of first-time buyers within the new-build market.

The paper yields three main findings. First, I document clear excess mass just below the €500,000 cutoff in the treated new-build market during both policy regimes, together with missing mass above it, whereas no comparable pattern appears either in the pre-policy period or in the second-hand market. Under the polynomial counterfactual, the normalised bunching coefficient is $\hat{b} = 1.352$ under the initial regime and 2.090 under the enhanced regime, implying a transaction-value response of €6,759 and €10,448 for the marginal developer who relocates below the threshold, around one-third of the maximum statutory relief in each regime. The bunching coefficient rises by 55% against a 50% rise in maximum statutory relief between regimes, in line with a local response of the threshold distortion that strengthens with subsidy generosity. An alternative counterfactual reported in Appendix B preserves the monotonic increase in $\hat{b}$ with subsidy generosity, though with a smaller cross-regime ratio.

Second, a distributional difference-in-differences shows that this response is highly localised around the threshold rather than diffuse across a wider range of transaction-values, unlike settings in which frictions can produce diffuse adjustment across a broader range (Anagol et al., 2024).

Third, the buyer-participation response is economically large and precisely estimated. Relative to second-hand purchases, the first-time buyer share of new-build purchases increases by 5.11 percentage points under the initial regime and by 10.50 percentage points under the enhanced regime. Relative to former owner-occupiers, the new-build share of first-time buyer purchases rises by 2.48 and 5.33 percentage points respectively. Across both designs, the estimated response scales closely with the doubling of the marginal subsidy rate from 5% to 10%, consistent with an approximately linear local buyer-participation response to subsidy generosity over the observed policy range.

This paper makes three contributions. First, it provides the first market-level analysis of Ireland's HTB scheme, complementing the borrower-level evidence in McCann and Singh (2023) by studying the transaction-value distribution and the cross-segment composition of market activity. Second, it extends the housing-notch literature by examining a partial-coverage notch rather than a universal one. Relative to settings such as Best and Kleven (2018) and Kopczuk and Munroe (2015), the Irish scheme affects only part of market demand, which both attenuates the transaction-value response and creates scope for changes in buyer-participation as an empirical object of interest. Third, it adds to a smaller literature on how policy and financing shocks reshape buyer composition in housing markets (Berger, Turner, and Zwick, 2020; Carozzi, 2020; Lambie-Hanson, Li, and Slonkosky, 2022), which has typically studied shocks that unintentionally displaced particular buyer groups, by documenting a substantial cross-segment reallocation of first-time buyer-participation from the second-hand market into new-builds under an explicitly targeted policy, and showing that this reallocation scales closely with subsidy generosity.

The remainder of the paper is organised as follows. Sections 2 and 3 describe the policy context and review the related literature. Section 4 develops the theoretical framework, and Section 5 describes the data. Section 6 sets out the empirical strategy, Section 7 presents the main results, and Section 8 examines robustness. Section 9 concludes.

2 Policy Context

Introduced in the national budget in October 2016, Ireland's Help to Buy (HTB) scheme has formally applied to qualifying purchases and self-builds from the 1st of January 2017 onwards. It also briefly allowed retrospective claims for qualifying transactions undertaken between the 19th of July 2016 and the 31st of December 2016[1]. HTB is paid at the deposit stage, after the signing of the contract in the case of a purchase, or following the drawdown of the first tranche of the mortgage in the case of a self-build. It consists of a repayment of income tax and Deposit Interest Retention Tax (DIRT)[2] paid over the four tax years preceding the application (Revenue Commissioners, 2024). This distinguishes it from the Help to Buy schemes in the United Kingdom, which took the form of government-backed equity or shared-equity loans (Department for Levelling Up and Communities, 2024; Welsh Government, 2026).

The scheme applies only to first-time buyers (FTBs) of new-build properties and is unavailable for properties with a transaction-value above €500,000. Second-hand properties have never been eligible, which makes them a natural placebo group in the empirical analysis. Eligibility also requires owner-occupation as a Principal Private Residence[3], a qualifying mortgage with a loan-to-value ratio of at least 70%, and full tax compliance (Revenue Commissioners, 2024).

Let T denote the amount of eligible tax paid over the relevant four-year period, and let

[1]These transactions were completed, and their transaction-values set, before the scheme was announced in the October 2016 budget. The retrospective eligibility window therefore does not affect the transaction-value data used in the bunching analysis.

[2]Deposit Interest Retention Tax (DIRT) is a tax on interest earned on bank and other deposit accounts in Ireland.

[3]A Principal Private Residence is a house or apartment which is owned and occupied as an individual's only, or main, residence.

v denote the qualifying transaction-value of the residence, or the approved valuation in the case of a self-build.

Under the original regime, which applied from the 1st of January 2017 to the 22nd of July 2020, relief was given by

$$\text{HTB}_{2017}(v,T) = \begin{cases} \min\{0.05v,\ €20{,}000,\ T\}, & \text{if } v \leq €500{,}000 \\ 0, & \text{if } v > €500{,}000. \end{cases}$$

The subsidy therefore equalled 5% of the transaction value, capped at €20,000, and could not exceed prior eligible tax paid. At $v = €400{,}000$, the subsidy reaches its maximum, so the marginal subsidy rate falls from 5% to 0%. The original schedule therefore contains a kink at €400,000 and a notch at $v = €500{,}000$, where the subsidy drops to zero.

A temporary enhancement was introduced on the 23rd of July 2020 and has since been extended to the 31st of December 2029. Under this enhanced regime, relief is given by

$$\text{HTB}_{2020}(v,T) = \begin{cases} \min\{0.10v,\ €30{,}000,\ T\}, & \text{if } v \leq €500{,}000 \\ 0, & \text{if } v > €500{,}000. \end{cases}$$

Under this version of the scheme, the subsidy is equal to 10% of the transaction value, capped at €30,000, and again cannot exceed prior eligible tax paid. At $v = €300{,}000$, the subsidy reaches its maximum, so the marginal subsidy rate falls from 10% to 0%. The enhanced schedule therefore also contains a kink, though at €300,000. At $v = €500{,}000$, the subsidy again drops to zero, creating the same notch in the policy schedule.

Figure 1 summarises the key non-linearities in the two HTB schedules. The empirical analysis focuses on the common notch[4], since eligibility is lost entirely at that point and the relevant pricing incentives change discretely.

[4] Although the HTB schedule also contains kinks at €400,000 and €300,000, I do not use a regression kink design at those thresholds, since analogous slope changes are also present in placebo exercises. This limits the credibility of a causal interpretation of the results.

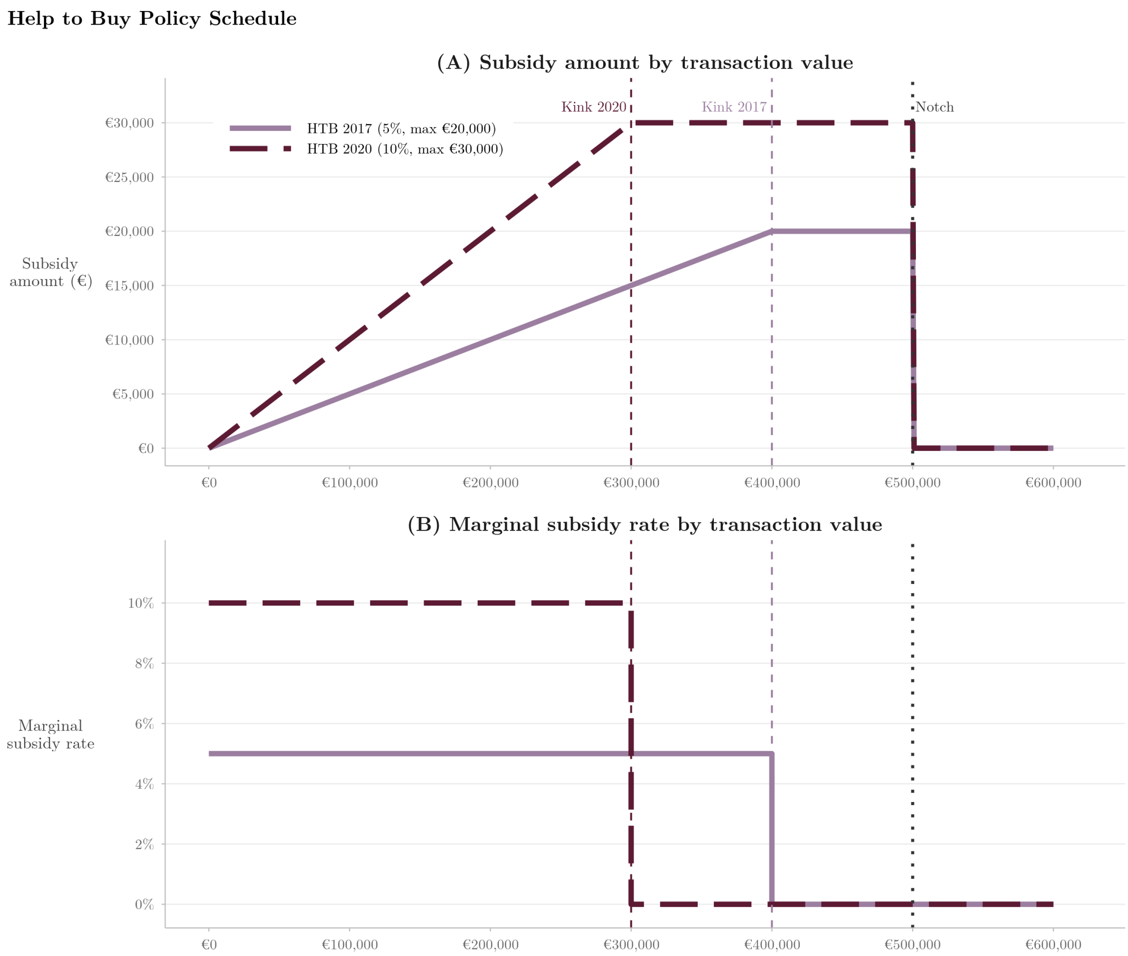

Figure 1: Help to Buy subsidy schedule by qualifying transaction-value.

HTB has operated at substantial scale since its introduction. From 2017 to 2022, approved claims for purchased new-build properties ranged from 3,900 to 5,500 per year (Revenue Commissioners, 2026). At the same time, the new-build market was not composed exclusively of first-time buyers: household former owner-occupiers still accounted for 3,000 to 3,900 such purchases per year over the same period (Central Statistics Office, 2026). Any market-level response at the €500,000 threshold is therefore observed in a segment that includes both HTB-eligible and HTB-ineligible demand.

The distribution of approved claims also moved towards higher-value properties over time, thereby increasing the relevance of the notch in the HTB schedule. In 2017, just over half of claims were for properties valued below €300,000, whereas by 2022 around 44% were for properties above €375,000, and 12.5% were in the €450,000–€500,000 band (Revenue Commissioners, 2026). Over the same period, the statutory cap became increasingly relevant: in the full years 2017–2019, 12 to 16% of claims were at the €20,000 maximum, whereas after the July 2020 enhancement the share at the

€30,000 cap rose from 21% in 2020 to 57% by 2022 (Revenue Commissioners, 2026). These patterns indicate that the upper end of the eligibility range, and in particular the €500,000 cutoff, became increasingly important over time.

3 Related Literature

A central result in the housing-tax literature is that non-linear transaction-tax schedules generate excess mass just below salient thresholds. Best and Kleven (2018) establish this result in the context of stamp-duty in the United Kingdom, showing substantial bunching below notches and corresponding missing mass above them, with transaction-values responding by a factor of two to five times the tax increase. Proportional responses are largest at the lower end of the price distribution, where buyer liquidity constraints are tighter and the tax jump is large relative to available resources. Evidence from other contexts reinforces this core finding. Slemrod, Weber, and Shan (2017) document bunching after transfer-tax reforms in Washington, D.C., with the more prominent 2006 reform generating stronger responses than the smaller 2003 reform. Kopczuk and Munroe (2015) show that sales in New York cluster below the mansion-tax threshold, whilst C. K. Y. Leung, T. C. Leung, and Tsang (2015) find that both the location and intensity of bunching move when the Hong Kong stamp-duty schedule changes. This evidence suggests that threshold responses depend on the salience, effective magnitude, and position of the notch in the price distribution. It is also consistent with the observation by Best and Kleven (2018) that housing-market participants appear less affected by optimisation frictions than agents in labour markets, which may help explain why such responses are often highly local. At the same time, Kopczuk and Munroe (2015) note that housing markets exhibit substantial round-number heaping even in the absence of policy thresholds, a point that is especially important in the present setting as the HTB ceiling lies at €500,000, an inherently salient price point.

Excess mass below a threshold is only part of the distortion. Kopczuk and Munroe (2015) find that the gap above the New York threshold exceeds the bunching mass below it, which they interpret as evidence that some mutually beneficial transactions fail to occur because the notch disrupts matching. Evidence on first-time buyer stamp-duty exemptions further illustrates the broader consequences of such thresholds. Hargaden (2017), studying the Irish pre-2007 stamp-duty exemption for first-time buyers, finds a double-bunching pattern in which first-time buyers bunch below the threshold and second-time buyers bunch above it, implying an allocative distortion, as above the threshold second-time buyers face a smaller tax increase and can therefore outbid first-time buyers more easily. Clemente-Casinhas and Vale (2026), studying Portugal's 2024 stamp-duty exemption for first-time buyers under 35, show that the exemption led to a significant increase in asking price growth, with the tax savings offset within 3 to 15 months, and that the response was strongest in supply-constrained cities and areas with more eligible buyers. These findings show that threshold-based housing tax policies can generate not only bunching, but also disrupted matching, allocative distortions, and price inflation or capitalisation.

The evidence on capitalisation also connects this threshold literature to a related literature on housing purchase assistance more broadly, where the central question is whether such policies improve affordability or are instead capitalised into prices. Krolage (2023), studying a Bavarian housing purchase subsidy, finds price increases for single-family homes consistent with full capitalisation, but no comparable effect for apartments whose buyers were less likely to qualify. Carozzi, Hilber, and Yu (2024), studying Britain's Help to Buy equity loan scheme, exploit discontinuities in scheme generosity at administrative boundaries and again show that incidence depends on local supply elasticity: in supply-constrained London the policy raised prices without increasing construction, whereas near the English-Welsh border it increased construction with little price effect. This literature therefore shows that supply conditions are central to subsidy incidence, but also that eligibility rules may shape the composition of the housing stock itself. Agarwal, Hu, and Lee (2023), studying a subsidy for new-build homes in New South Wales, Australia, find strong bunching for policy-eligible homes, no comparable pattern for ineligible second-hand homes, substantial capitalisation of the relief into prices, and evidence that eligible homes are smaller and sell more quickly. Uribe (2022), studying Colombia's low-cost housing policy, shows that greater subsidy generosity increases bunching at the eligibility threshold and induces households to reduce housing expenditure, which in equilibrium translates into smaller units. Once eligibility depends on a price ceiling, the relevant adjustment margins may extend beyond prices to include housing characteristics and the composition of the stock.

This paper also relates to a smaller literature on how policy and financing shocks affect participation patterns in housing markets. Carozzi (2020) shows that tighter credit conditions in the United Kingdom disproportionately reduced first-time buyer transactions and pulled sales away from the lower end of the market. Berger, Turner, and Zwick (2020) show that the US First-Time Homebuyer Credit concentrated its effects in the existing-home and starter-home segments rather than in new construction, and that the age distribution of buyers moved towards younger, downpayment-constrained households during the policy period. Lambie-Hanson, Li, and Slonkosky (2022) similarly show that, during the US housing recovery, investors gained market share in housing transactions as mortgage credit supply tightened for individual buyers. These papers indicate that policy and financing shocks may reshape not only prices and quantities, but also which buyers transact and in which market segment.

To my knowledge, the only existing study of Ireland's HTB scheme is McCann and Singh (2023), who show that the dominant borrower-level response to the July 2020 enhancement operated through liquidity: eligible first-time buyers reduced their out-of-pocket downpayments by approximately €10,000, the amount of the additional subsidy, with limited effects on leverage or purchase prices, and no detectable change in the characteristics of eligible first-time buyers across the enhancement. Their design does not, however, examine the €500,000 eligibility ceiling or the market-level distribution of transaction-values, nor whether HTB tilted buyer-participation in the new-build market towards first-time buyers. The Irish HTB scheme is well suited to studying these margins because it combines a housing-purchase subsidy with a salient transaction-value cutoff, whilst being delivered through the tax system rather than through a grant or equity-style support. This makes it possible to examine both the threshold distortions generated by the eligibility ceiling and the changes in buyer-participation induced by the subsidy within a single policy setting.

4 Theoretical Framework

In the Irish new-build market, transaction-values are largely set by developers rather than determined through an open bidding process[5]. Developers therefore have local discretion around the threshold and are the relevant bunching agent. I accordingly model the developer as choosing the transaction-value of a unit in the presence of a discontinuity in demand at $\bar{v}$ = €500,000. Crossing the threshold does not merely increase the buyer's gross transaction value, but removes access to HTB for eligible purchasers, reducing effective willingness to pay and lowering the probability of sale. Some developers whose latent preferred transaction-values lie just above $\bar{v}$ may therefore find it more profitable to locate at or just below the cutoff, because preserving HTB eligibility raises the probability of sale enough to outweigh the reduction in net revenue per sale associated with bunching below the threshold.

[5]The Irish Competition and Consumer Protection Commission (2025) notes that, in the case of a new-build home, "the purchase price is usually set in advance and consumers do not engage in a bidding process".

4.1 Demand

Following Polyakova and Ryan (2019), I model demand as the aggregation of demands from heterogeneous buyers facing a uniform seller-chosen transaction value, with targeted subsidies entering through the net transaction-value paid by eligible consumers. As in Best and Kleven (2018), the analysis is conducted in terms of transaction-values $v = p \cdot h$, where p denotes the price of one unit of housing quality and h the quality-adjusted quantity of housing services. Since HTB eligibility depends on v rather than on p or h separately, the model treats transaction-value as the relevant choice object and abstracts from separately modelling the unit price and housing characteristics. A reduction in v should therefore be interpreted as a reduction in transaction value, which may reflect an adjustment in the unit price, housing characteristics, or both.

Buyers are heterogeneous in willingness to pay and HTB eligibility. Since a transaction-value of $v = \bar{v}$ remains HTB-eligible, I define demand at the threshold by the left-hand limit, $D(\bar{v}) \equiv D(\bar{v}^-)$. Let $\lambda \in (0,1)$ denote the effective share of buyers in the relevant new-build market segment who are HTB-eligible first-time buyers. Let $D_I(v)$ be the purchase probability of an ineligible buyer at transaction-value v, and let $D_E(x)$ be the purchase probability of an eligible buyer when their effective transaction-value is x. Assume D_I and D_E are continuous and strictly decreasing in their arguments.

Let $\tau > 0$ denote the effective value of HTB eligibility near $\bar{v}$, interpreted as the reduction in the effective transaction-value faced by an eligible buyer at the threshold. Eligible buyers face an effective transaction-value of $v - \tau$ for $v \leq \bar{v}$ and v for $v > \bar{v}$. The resulting reduced-form sale probability faced by the developer is therefore[6]

$$D(v) = \begin{cases} (1-\lambda)\, D_I(v) + \lambda\, D_E(v-\tau) & v \leq \bar{v}, \\ (1-\lambda)\, D_I(v) + \lambda\, D_E(v) & v > \bar{v}. \end{cases} \tag{1}$$

At $\bar{v}$ there is a discrete downward jump in demand of size

$$\Delta D \equiv D(\bar{v}^-) - D(\bar{v}^+) = \lambda\big[D_E(\bar{v}-\tau) - D_E(\bar{v})\big] > 0, \tag{2}$$

which is positive because D_E is strictly decreasing and $\tau > 0$. It is convenient to define

$$E(\tau, \lambda) \;\equiv\; \lambda\big[D_E(\bar{v}-\tau) - D_E(\bar{v})\big], \tag{3}$$

[6] In the spirit of Polyakova and Ryan (2019), one could write aggregate demand more generally as $D(v) = \int q(v - s(v,\theta), \theta)\; dF(\theta)$, where θ indexes buyer type and $s(v,\theta)$ denotes the subsidy received by type θ at transaction-value v. The specification used in this paper is the discrete two-group special case of this general structure.

which I refer to as effective exposed demand at the notch. This is the share of demand that is sensitive to losing eligibility above the cutoff, weighted by the value of that eligibility. By construction, $\Delta D = E(\tau, \lambda)$, so the size of the demand discontinuity is fully summarised by this single object. The two inputs to $E(\tau, \lambda)$, namely the value of eligibility τ and the eligible share of demand λ, enter the threshold distortion symmetrically, and any change in subsidy generosity that raises either input strengthens the notch.

The partial derivatives of $E(\tau, \lambda)$ make the role of each input explicit. Holding τ fixed,

$$\frac{\partial E(\tau, \lambda)}{\partial \lambda} = D_E(\bar{v} - \tau) - D_E(\bar{v}) > 0,$$

so effective exposed demand is larger when the relevant new-build segment contains a higher share of HTB-eligible first-time buyers. Holding λ fixed, if D_E is differentiable, then

$$\frac{\partial E(\tau, \lambda)}{\partial \tau} = -\lambda D_E'(\bar{v} - \tau) > 0,$$

so effective exposed demand is also larger when the value of eligibility rises. Both the generosity of the subsidy and the composition of participating buyers therefore matter for the size of the notch-induced decrease in sale probability faced by developers, and they do so through the same channel.

If, in addition, the policy itself draws more first-time buyers into the new-build market, then λ is endogenous to τ. Considering $\lambda = \lambda(\tau)$ with $\lambda'(\tau) > 0$, the total derivative of effective exposed demand with respect to subsidy generosity becomes

$$\frac{dE(\tau, \lambda)}{d\tau} = \underbrace{\lambda'(\tau)\big[D_E(\bar{v} - \tau) - D_E(\bar{v})\big]}_{\text{participation}} + \underbrace{\big(-\lambda(\tau) D_E'(\bar{v} - \tau)\big)}_{\text{direct valuation}} > 0.$$

The first term captures the participation channel: a more generous subsidy increases the eligible share of demand. The second captures the direct valuation channel: each eligible buyer values eligibility more. Both enlarge effective exposed demand at the threshold, and therefore strengthen the developer's incentive to bunch through a single mechanism with two reinforcing inputs.

4.2 Developer's Problem

The developer is modelled as choosing the transaction-value of a single new-build unit. This captures the relevant local transaction-value decision near the threshold. Let $S(v) \in \{0, 1\}$ denote an indicator for whether the unit sells at transaction-value

v. Realised profit is therefore

$$\Pi(v) = (v - c)S(v),$$

where c is the developer's marginal cost of supplying the unit, interpreted in reduced form as the opportunity cost associated with that transaction. Taking expectations conditional on v yields

$$\pi(v) \equiv \mathbb{E}[\Pi(v) \mid v] = (v - c)\Pr(S(v) = 1 \mid v).$$

Since $D(v)$ denotes the sale probability implied by the demand system, profit can be written as

$$\pi(v) = (v - c)D(v). \tag{4}$$

Thus, the developer chooses the transaction-value v to maximise expected profit, trading off net revenue per sale, $(v - c)$, against the probability of sale. This is consistent with the housing literature, in which sellers face a trade-off between the value at which a property is offered and the probability of sale (Guren, 2018). In the present setting, the HTB scheme enters through $D(v)$. The discontinuity established in (2) then implies that, at the cutoff,

$$\pi(\bar{v}^-) = (\bar{v} - c)D(\bar{v}^-), \qquad \pi(\bar{v}^+) = (\bar{v} - c)D(\bar{v}^+),$$

so

$$\pi(\bar{v}^-) - \pi(\bar{v}^+) = (\bar{v} - c)\big[D(\bar{v}^-) - D(\bar{v}^+)\big] = (\bar{v} - c)\, E(\tau, \lambda) > 0$$

whenever $c < \bar{v}$. This condition holds for any developer whose marginal cost is below the threshold, which is necessary for the unit to be profitably supplied in this transaction-value range. Moreover,

$$\begin{aligned}\frac{d}{d\tau}\big[\pi(\bar{v}^-) - \pi(\bar{v}^+)\big] &= (\bar{v} - c)\frac{dE(\tau, \lambda)}{d\tau} \\ &= (\bar{v} - c)\Big[\lambda'(\tau)\big(D_E(\bar{v} - \tau) - D_E(\bar{v})\big) - \lambda(\tau)D_E'(\bar{v} - \tau)\Big] > 0,\end{aligned}$$

so the profit gain from locating at or below the threshold is larger when subsidy generosity rises. The incentive to bunch therefore strengthens with effective exposed demand $E(\tau, \lambda)$, which is itself increasing in both the direct value of eligibility and, through the endogenous participation channel, the eligible share of demand.

4.3 Counterfactual Distribution

To connect the individual transaction-value-setting problem to the observed distribution of transaction-values, I now allow developers to differ in marginal cost. Following the bunching literature, let $f(v)$ denote the smooth counterfactual density of transaction-values that would prevail absent the notch at $\bar{v}$ (Kleven, 2016; Best and Kleven, 2018). The key assumption is that, in the absence of the notch, the density of transaction-values is smooth in a neighbourhood of $\bar{v}$.

Suppose developers differ in marginal cost c, with smooth density $g(c)$. In the no-notch environment, a developer of type c chooses transaction-value v to maximise

$$\pi^0(v; c) = (v - c)D^0(v),$$

where $D^0(v)$ denotes the smooth sale-probability schedule that would prevail absent the discontinuity at $\bar{v}$. Let $v^0(c)$ denote the corresponding optimal transaction-value. If the mapping $c \mapsto v^0(c)$ is monotone, it induces a counterfactual density of latent transaction-values

$$f(v) = g\big((v^0)^{-1}(v)\big) \left| \frac{d}{dv}(v^0)^{-1}(v) \right|. \tag{5}$$

The smooth density $f(v)$ used in the bunching analysis can therefore be interpreted as the density of latent preferred transaction-values generated by heterogeneous developers in the absence of the notch.

4.4 Bunching

As in Sallee and Slemrod (2012), the developer's transaction-value-setting problem can be interpreted as a two-stage decision. Let v^0 denote the transaction-value implied by earlier development and specification choices in the no-notch benchmark. Close to sale, the developer may still adjust the final transaction-value in response to proximity to the HTB threshold. Bunching at $\bar{v}$ should therefore be understood as a local adjustment in transaction value, which may be achieved through changes in the unit price, housing characteristics, or both.

Let $\Delta v^* > 0$ denote the upper bound of the latent transaction-value interval for which bunching at $\bar{v}$ is optimal. In the frictionless homogeneous benchmark, the observed transaction-value rule under the notch is

$$v^*(v^0) = \begin{cases} v^0, & v^0 \le \bar{v}, \\ \bar{v}, & v^0 \in (\bar{v}, \bar{v} + \Delta v^*), \\ v^0, & v^0 > \bar{v} + \Delta v^*. \end{cases} \tag{6}$$

Thus, the notch leaves transaction-values unchanged outside the manipulated region but reallocates the counterfactual mass from $(\bar{v}, \bar{v} + \Delta v^*)$ to the single point $\bar{v}$.

The marginal buncher is the developer type c^* whose latent preferred transaction-value lies exactly at the upper boundary of the bunching region:

$$v^0(c^*) = \bar{v} + \Delta v^*.$$

Following the standard notch logic of Kleven and Waseem (2013), type c^* is indifferent between bunching at $\bar{v}$ and choosing its latent preferred transaction-value $v^0(c^*)$, so

$$(\bar{v} - c^*)D(\bar{v}) = \big(v^0(c^*) - c^*\big)D\big(v^0(c^*)\big).$$

Substituting $v^0(c^*) = \bar{v} + \Delta v^*$ gives

$$(\bar{v} - c^*)D(\bar{v}) = (\bar{v} + \Delta v^* - c^*)D(\bar{v} + \Delta v^*). \tag{7}$$

Equation (7) characterises the homogeneous benchmark in which bunchers share a common upper bound Δv^*.

4.5 Dominated Region

Because expected profit falls discretely at $\bar{v}$, there exists a neighbourhood above the cutoff in which choosing an interior transaction-value is strictly dominated by bunching at the threshold. More precisely, for each developer type c with $c < \bar{v}$, the profit gap

$$\pi(\bar{v}; c) - \pi(\bar{v}^+; c) = (\bar{v} - c)\, E(\tau, \lambda)$$

is strictly positive. Since $\pi(v; c)$ is continuous for $v > \bar{v}$, there exists for each such type c an interval

$$(\bar{v}, \bar{v} + \Delta v^D(c))$$

such that

$$\pi(\bar{v}; c) > \pi(v; c) \qquad \text{for all } v \in (\bar{v}, \bar{v} + \Delta v^D(c)).$$

Hence, for developer type c, choosing a transaction-value slightly above the threshold is strictly dominated by bunching at the cutoff. At the upper boundary $\bar{v} + \Delta v^D(c)$, the developer is indifferent between bunching at the cutoff and choosing the interior transaction value:

$$(\bar{v} - c)D(\bar{v}) = (\bar{v} + \Delta v^D(c) - c)\, D(\bar{v} + \Delta v^D(c)).$$

Thus, in the frictionless benchmark, each bunching type has a locally dominated region above the cutoff. Aggregating across developer types therefore yields excess bunching at $\bar{v}$ together with missing mass above the threshold, consistent with standard notch predictions (Kleven and Waseem, 2013; Best and Kleven, 2018). The frictionless benchmark is useful as a reference case, although the empirical response need not take the literal form of a perfect spike at $\bar{v}$ and zero mass immediately above it.[7]

4.6 Implication for Bunching Mass

Following the logic of Best and Kleven (2018), consider the homogeneous frictionless benchmark in which there is a single bunching segment $(\bar{v}, \bar{v} + \Delta v^*)$. Let $B(\bar{v})$ denote the excess bunching mass at the threshold, and let $M(\bar{v})$ denote the missing mass above it. In what follows, the key object is $B(\bar{v})$, the amount of counterfactual mass reallocated to the cutoff from transaction-values just above $\bar{v}$. With $f(v)$ the counterfactual density of transaction-values absent the notch, then

$$B(\bar{v}) = \int_{\bar{v}}^{\bar{v}+\Delta v^*} f(v)\, dv. \tag{8}$$

The corresponding missing mass is the reduction in density above the cutoff induced by bunching, but the mapping from bunching to the implied transaction-value response relies on $B(\bar{v})$.

If the counterfactual density is approximately flat in a neighbourhood of the notch, (8) simplifies to

$$B(\bar{v}) \approx f(\bar{v})\, \Delta v^*. \tag{9}$$

Rearranging gives the implied transaction-value response:

$$\Delta v^* \approx \frac{B(\bar{v})}{f(\bar{v})}. \tag{10}$$

[7] Developers may retain limited flexibility over transaction-values close to sale, may revise later-phase units as information about demand is revealed, and may still face some untargeted demand from ineligible buyers above the threshold. More generally, notches need not generate zero density above the cutoff under frictions or sparsity in feasible opportunities (Anagol et al., 2024).

Under local flatness, excess bunching therefore identifies the transaction-value response of the marginal buncher in the homogeneous case[8].

Predictions 1. Suppose $\Delta D > 0$ and the counterfactual density $f(v)$ is smooth and strictly positive in a neighbourhood of $\bar{v}$. This theoretical framework delivers four empirical predictions:

(i) **Excess mass at or just below the notch**: in the HTB-treated new-build market, developers with latent preferred transaction-values just above $\bar{v}$ will relocate to the cutoff, generating excess mass at or immediately below the threshold together with reduced density above it.

(ii) **Stronger responses under greater subsidy generosity**: the response should be stronger when effective exposed demand $E(\tau, \lambda)$ is larger, which occurs when either the value of eligibility τ or the eligible share of demand λ rises. Empirically, this implies larger distortions under the enhanced regime, which raises the maximum benefit at the threshold from €20,000 to €30,000.

(iii) **Positive implied transaction-value response**: the excess bunching mass at the threshold, together with the smooth counterfactual density at $\bar{v}$, identifies the transaction-value response of the marginal buncher,

$$\Delta v^* \approx \frac{B(\bar{v})}{f(\bar{v})},$$

in the homogeneous benchmark, or a local average transaction-value response under heterogeneity.

(iv) **Buyer-participation**: if HTB increases the relative attractiveness of eligible new-build purchases for first-time buyers, the effective share of eligible demand in the relevant new-build segment, λ, should rise after the policy is introduced.

5 Data

This section describes the two administrative datasets used in the analysis and how they relate to the paper's empirical strategy. The Property Price Register is used to study transaction-values and threshold responses around €500,000, whilst the Central Statistics Office series on market-based household purchases is used to analyse buyer-participation in the new-build market. I then present descriptive statistics for both

[8] Equation (9) implicitly assumes a single bunching segment and hence a homogeneous population of bunchers. More generally, if developer type θ has transaction-value response $\Delta v(\theta)$ and type-specific counterfactual density $\tilde{f}(v, \theta)$, then $B(\bar{v}) = \int_\theta \int_{\bar{v}}^{\bar{v}+\Delta v(\theta)} \tilde{f}(v, \theta)\, dv\, d\theta \approx f(\bar{v})\, \mathbb{E}[\Delta v]$, hence bunching identifies a local average transaction-value response across types near the threshold.

sources.

5.1 Data Description

5.1.1 Property Price Register

The main dataset used in this paper is the Property Price Register (PPR). The PPR is an administrative dataset containing the date of sale, transaction value, property type, and address of all arm's length residential property sales in Ireland since the 1st of January 2010 (Property Services Regulatory Authority, 2026). In particular, it records whether a property is classified as a new dwelling or as second-hand, which is central to the empirical strategy developed in this analysis. For new properties, the PPR reports transaction-values exclusive of VAT at 13.5% (Property Services Regulatory Authority, 2026). Because HTB eligibility and subsidy amounts are defined with reference to the gross transaction-value paid by the purchaser, I adjust new-build transaction-values to be VAT-inclusive throughout the analysis. The register is constructed from stamp-duty filings and, in principle, should therefore capture the near-universe of residential sales.

Some transactions are exempt from stamp-duty, or attract it only in reduced form, including certain transfers between spouses or civil partners, transfers following divorce, purchases under the local authority tenant purchase scheme, and certain transfers following a Property Adjustment Order (Citizens Information, 2025). These are unlikely to materially affect the analysis, since they do not constitute standard market transactions. The PPR can therefore be treated as a close approximation to the universe of residential market sales in Ireland.

The estimation sample is restricted to transactions that took place at their full market value[9], between the 1st of January 2013 and the 6th of July 2022. This start date excludes the immediate post-crisis period, during which Irish residential construction and new-build transactions remained severely depressed following the financial crisis and recession. These years are less representative of the market environment relevant for the later HTB period. The end date is chosen to avoid contamination from the introduction of the First Home Scheme on the 7th of July 2022, a government-backed shared-equity scheme for new-build purchases that overlaps with HTB and changes the effective set of supports available to first-time buyers.

The analysis is organised around six estimation cells, defined by two property types and three policy regimes. The two property types are new-build and second-hand

[9] This excludes transfers such as gifts, related-party sales, and other non-arm's-length transactions that do not reflect market transaction-values.

properties. The three policy regimes are the pre-policy period, running from the 1st of January 2013 to the 31st of December 2016, the initial regime, running from the 1st of January 2017 to the 22nd of July 2020, and the enhanced regime, running from the 23rd of July 2020 to the 6th of July 2022.

A limitation of the PPR is that it contains no hedonic property characteristics such as floor area, number of rooms, or building quality. This makes standard hedonic analysis infeasible in the present setting. Rather than attempting to estimate quality-adjusted price effects directly, the empirical strategy in this paper exploits the discontinuous incentives created by the HTB threshold and studies bunching in the transaction-value distribution.

5.1.2 Market-based Household Purchases of Residential Dwellings

To analyse buyer-participation patterns, I also use the Central Statistics Office series *Market-based Household Purchases of Residential Dwellings* (Central Statistics Office, 2026). This is a monthly administrative dataset, also derived from stamp-duty returns, recording the number of executed market purchases of residential dwellings by household buyers[10] by dwelling status, Residential Property Price Index regions[11], and buyer status. The dwelling-status classification distinguishes new-builds from second-hand dwellings, and buyer type is recorded separately for first-time buyer owner-occupiers, former owner-occupiers, and non-occupier household buyers.

I use the volume-of-sales series based on stamp-duty executions and restrict the sample to a similar period to the PPR, namely January 2013 to the end of June 2022. The series is therefore conceptually close to the PPR in coverage, but it is observed in aggregated rather than transaction-level form.

The principal advantage of this series is that it records buyer type, which is not observed in the PPR. Whilst the PPR is used to study transaction-values and bunching around the €500,000 threshold, this data makes it possible to analyse how residential purchases are distributed across treated first-time buyers and untreated former owner-occupiers, as well as other household buyers. Observed at the RPPI-region-by-month level, the data is used to study changes in the composition of participating buyers across dwelling status and buyer type over time.

[10] Here, household buyers refer to private individuals rather than non-household purchasers such as companies, charitable organisations, or state institutions.

[11] The RPPI regional series is slightly finer than the associated county-level series, since it separates the main cities from their surrounding counties.

5.2 Descriptive Statistics

5.2.1 Property Price Register

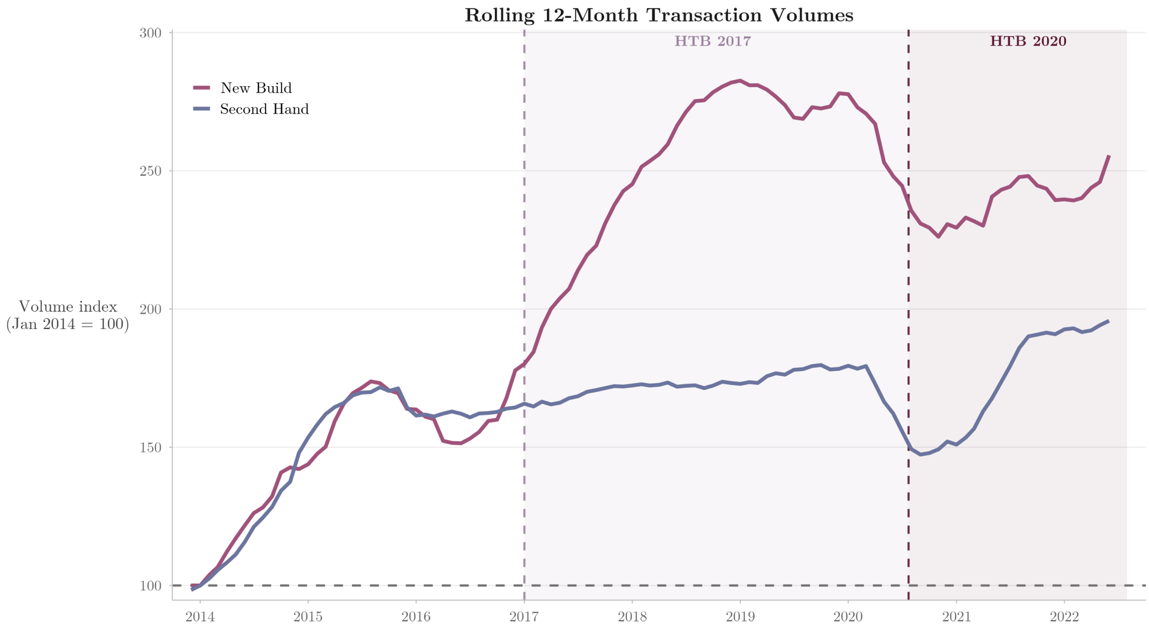

Figure 2: Rolling 12-month transaction volumes for new-build and second-hand properties, with January 2014 normalised to 100. Sample: January 2013 to June 2022.

Figure 2 plots rolling 12-month transaction volumes separately for new-build and second-hand properties. New-build volumes rise much more strongly than second-hand volumes over the sample period, with both series declining sharply in 2020 and 2021 as Covid-19 restrictions disrupted market activity before both begin to recover in 2022.

To summarise the full sample more formally, Table 1 reports descriptive statistics by regime and property type.

Table 1: Summary Statistics by Regime and Property Type

Regime	Type	Mean	Median	Mode	SD	N
Pre-Policy	New Build	€255,000	€211,000	€147,000	€183,000	20,275
Pre-Policy	Second Hand	€230,000	€175,000	€77,000	€196,000	131,536
HTB 2017	New Build	€358,000	€330,000	€331,000	€180,000	33,247
HTB 2017	Second Hand	€270,000	€220,000	€170,000	€210,000	142,099
HTB 2020	New Build	€395,000	€355,000	€333,000	€199,000	17,971
HTB 2020	Second Hand	€304,000	€250,000	€183,000	€219,000	89,289

Notes: The sample is restricted to sales with a gross transaction-value between €50,000 and €2,000,000 to exclude bulk or portfolio sales and likely non-arm's-length transfers misclassified in the PPR. All transaction-value statistics are rounded to the nearest €1,000.

Table 1 highlights three main features of the data. First, second-hand transactions substantially outnumber new-build transactions in every regime: there are 131,536 second-hand and 20,275 new-build transactions in the pre-policy period, 142,099 and 33,247 under the initial regime, and 89,289 and 17,971 under the enhanced regime. Second, mean transaction-values rise over time for both property types, but especially for new-builds, increasing from €255,000 in the pre-policy period to €358,000 under the initial regime and €395,000 under the enhanced regime. Median new-build transaction-values follow the same pattern, rising from €211,000 to €330,000 and then to €355,000. The modal new-build transaction-value also rises across regimes, from €147,000 in the pre-policy period to €331,000 and €333,000 in the two HTB regimes. This makes the €500,000 threshold increasingly relevant in later policy periods. Third, transaction-values remain widely dispersed in all cells. Standard deviations range from about €180,000 to €219,000, which motivates focusing later on a narrower estimation window around the threshold.

Because the bunching analysis is conducted in a restricted transaction-value window around the HTB threshold, Table 2 reports summary statistics for the estimation sample, €200,000–€600,000.

Table 2: Summary Statistics: Estimation Window

Regime	Type	Mean	Median	Mode	SD	N
Pre-Policy	New Build	€327,000	€305,000	€278,000	€94,000	9,548
Pre-Policy	Second Hand	€314,000	€285,000	€224,000	€97,000	50,419
HTB 2017	New Build	€351,000	€337,000	€334,000	€90,000	26,445
HTB 2017	Second Hand	€318,000	€292,000	€226,000	€97,000	71,972
HTB 2020	New Build	€360,000	€350,000	€336,000	€89,000	15,366
HTB 2020	Second Hand	€329,000	€305,000	€230,000	€99,000	51,888

Notes: The sample is restricted to sales with a gross transaction-value between €200,000 and €600,000. All transaction-value statistics are rounded to the nearest €1,000.

Table 2 shows that the restricted estimation window retains a substantial number of transactions in both treated and placebo groups. The broad upward movement in new-build transaction-values remains visible within the window, but it is more muted than in the full sample. At the same time, the restriction to €200,000–€600,000 removes much of the lower-valued and upper-tailed variation in the raw data, making the treated and placebo distributions more comparable in the part of the market relevant for identification. Dispersion falls substantially relative to the full sample, with standard deviations now ranging from €89,000 to €99,000. The window therefore preserves the local neighbourhood of the €500,000 threshold whilst discarding variation that is less informative for the bunching analysis.

Figures 3 and 4 present descriptive evidence on the transaction-value distributions around the €500,000 threshold. Figure 3 plots raw histograms in €5,000 bins by regime and property type, whilst Figure 4 plots local polynomial density estimates on either side of the cutoff.

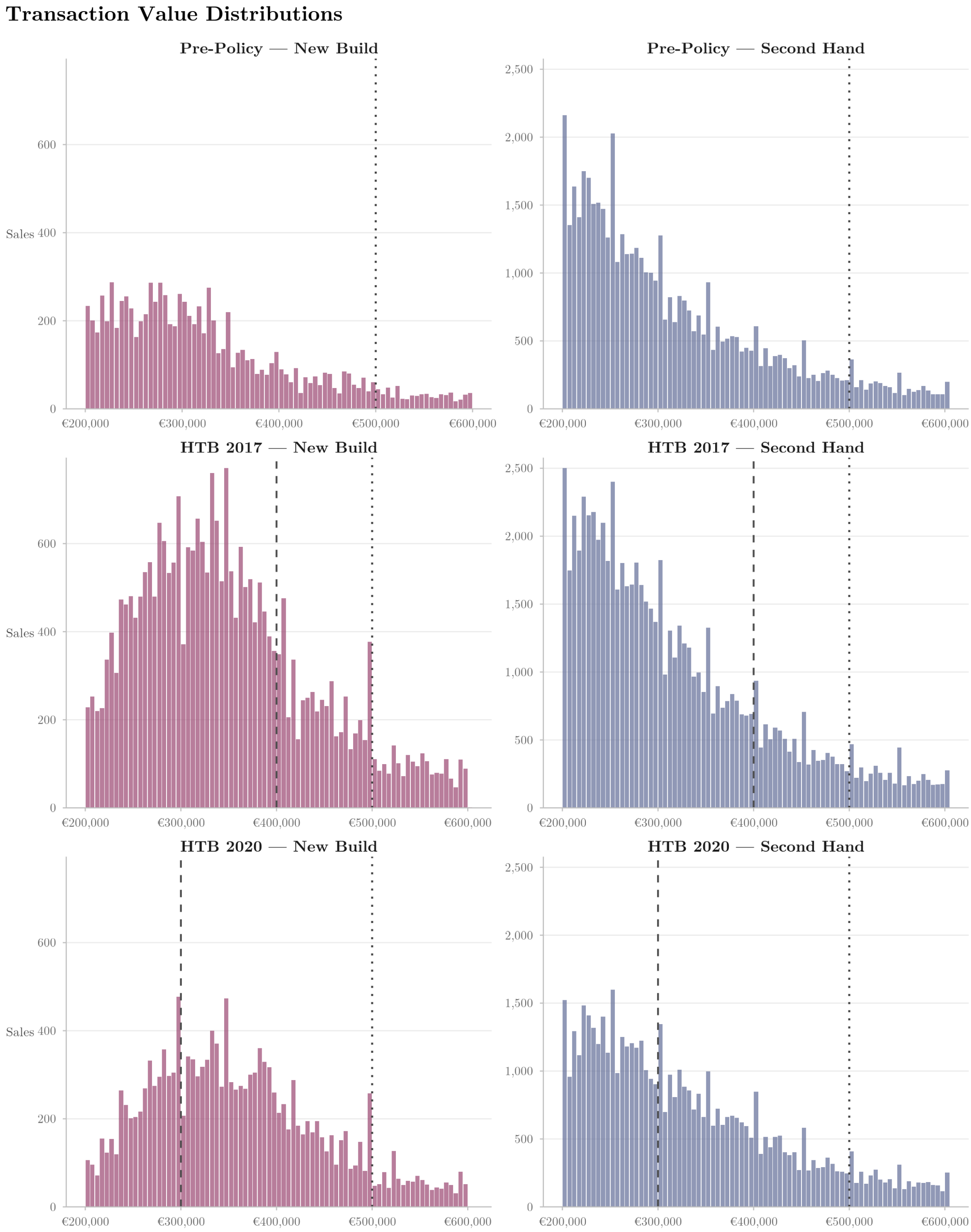

Figure 3: €5,000 bins, by property type and regime. Dashed lines indicate regime-specific kinks and the dotted line marks the €500,000 notch.

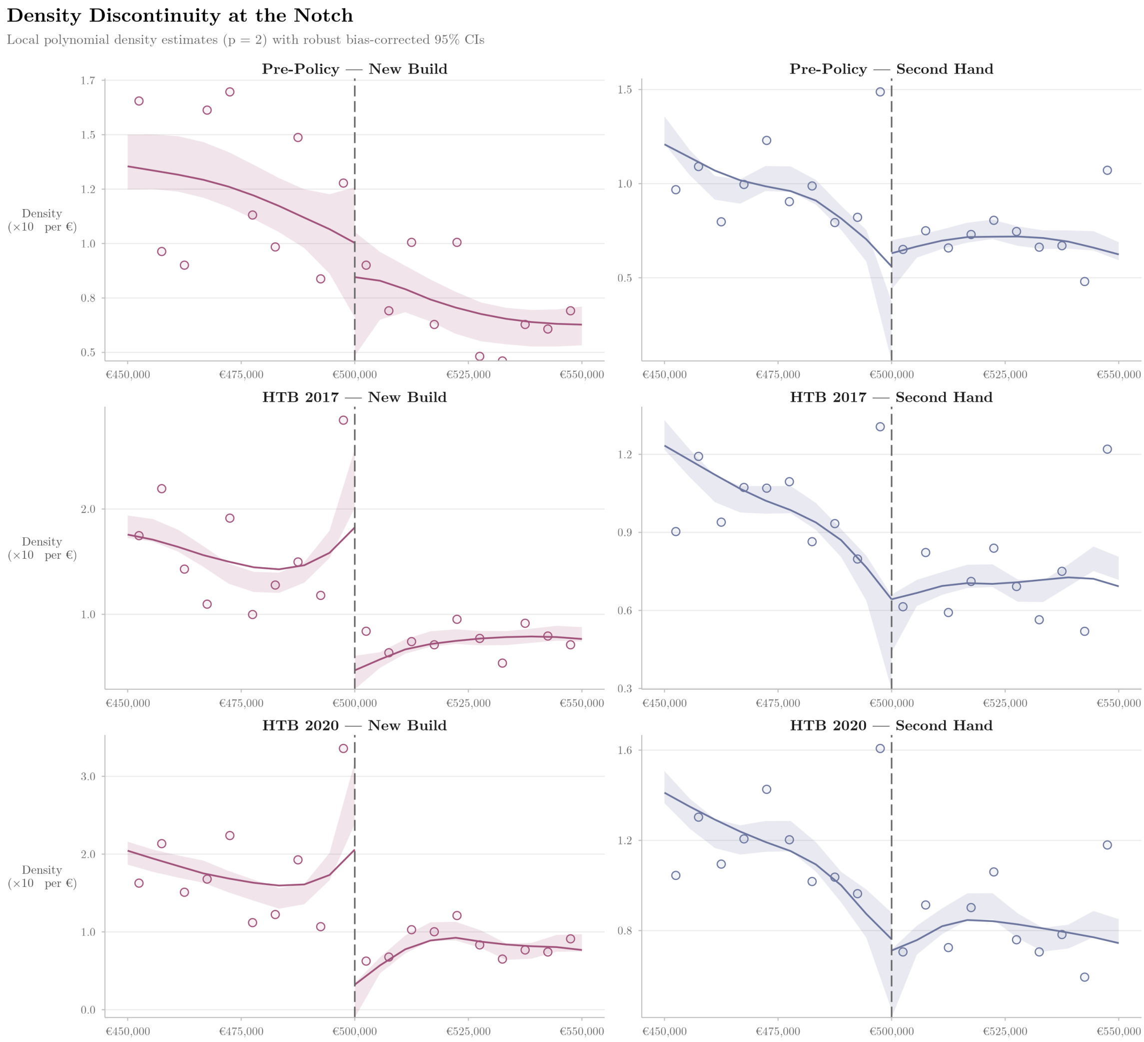

Figure 4: Local polynomial density estimates and 95% confidence intervals. Confidence intervals are robust and bias-corrected, and are therefore not necessarily centred around the point estimate. See Cattaneo, Jansson, and Ma (2018) for details.

Both figures point to a consistent pattern. In the pre-policy new-build distribution, there is no visible bunching at €500,000 and no statistically significant density discontinuity at the cutoff. Under both policy regimes, by contrast, new-build transactions bunch clearly in the €495,000–€500,000 bin immediately below the eligibility cutoff, and the local polynomial estimates exhibit large negative and statistically significant breaks at €500,000 in both the raw and jittered specifications. In the second-hand panels, the limited excess mass instead appears in the €500,000–€505,000 bin, including a small amount in the pre-policy period, and the only statistically significant density discontinuity appears in the raw pre-policy sample and disappears once transaction-values are jittered. Second-hand estimates are otherwise statistically insignificant in both policy periods. Since Kopczuk and Munroe (2015) note that round-number heaping in housing markets tends to occur in the bin above the focal value, the visual and formal evidence together point to policy-induced bunching in new builds rather than simple round-number heaping. The corresponding formal density estimates are reported in Appendix Table 11.

5.2.2 Market-based Household Purchases of Residential Dwellings

Figure 5 plots rolling 12-month transaction volumes for first-time buyers, former owner-occupiers, and non-occupier household buyers, separately for new-build and second-hand dwellings. In the new-build segment, all three buyer groups grow during the pre-policy recovery, but their paths diverge over time. Former owner-occupiers grow fastest in the early recovery period, but first-time buyers accelerate after 2017 and overtake them by 2019. Non-occupier household buyers peak around 2015 and then decline steadily, falling below their baseline during the enhanced regime. In the second-hand market, the pattern is less differentiated: first-time buyers and former owner-occupiers move broadly in parallel over the sample, whilst non-occupier household buyers again peak earlier and trend downwards during the period of study. The divergence between first-time buyers and other buyer types is therefore concentrated in the new-build segment, which is consistent with a stronger rise in first-time buyer-participation in that segment.

Table 3 reports descriptive totals from the data by policy regime, dwelling status, and buyer type. Two patterns are clear. First, second-hand purchases greatly exceed new-build purchases in every regime. Second, the buyer mix differs markedly across dwelling status. In the second-hand market, former owner-occupiers are the largest group throughout, accounting for 51.2% of purchases in the pre-policy period, 53.8% under the initial regime, and 56.1% under the enhanced regime. First-time buyers account for 25.1%, 26.7%, and 27.9% respectively, whilst non-occupier household buyers account

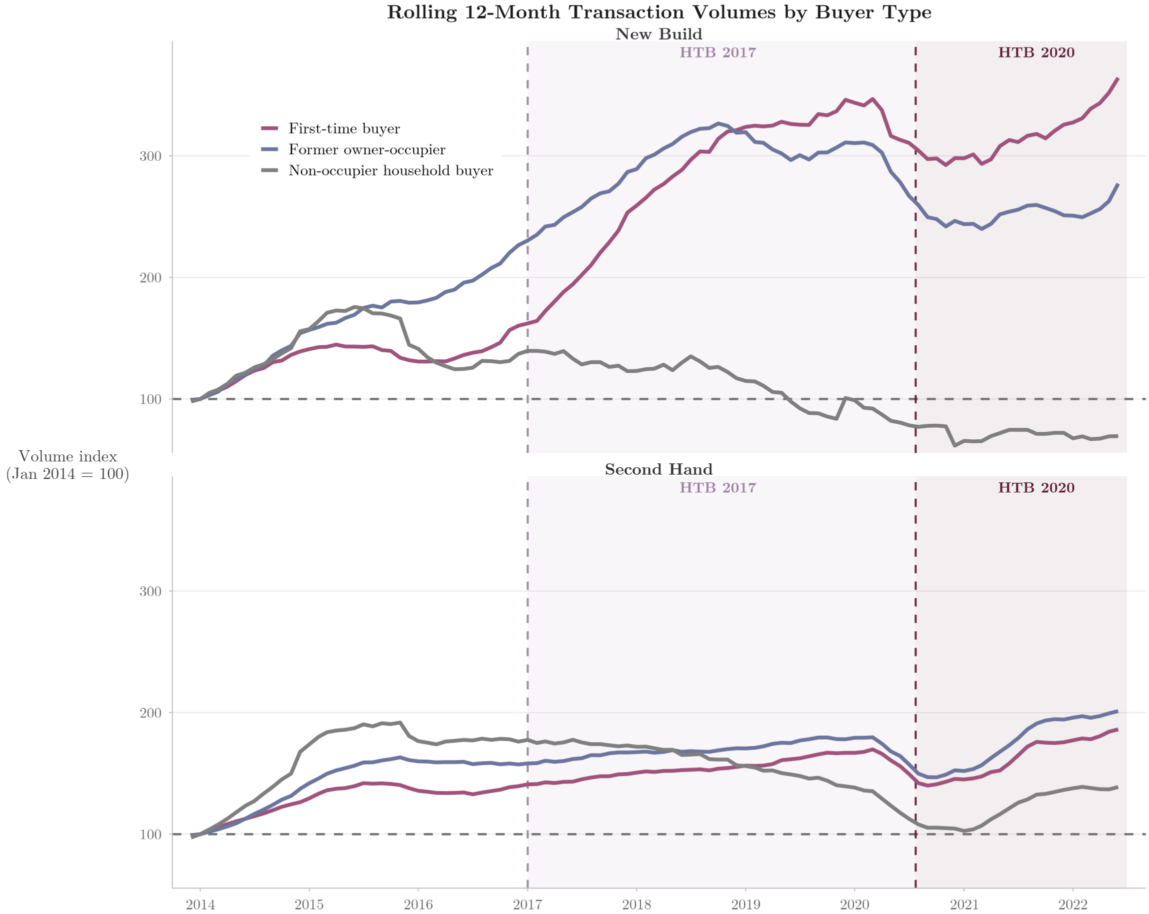

Figure 5: Rolling 12-month transaction volumes by buyer type and dwelling status, with January 2014 normalised to 100. Sample: January 2013 to June 2022.

Table 3: Market-Based Household Purchases by Dwelling Status, Buyer Type, and Regime

		Pre-Policy	HTB 2017	HTB 2020
New Build	First-time buyer	6,733	13,393	8,238
	Former owner-occupier	8,045	12,585	6,221
	Non-occupier household buyer	2,596	1,779	675
Second Hand	First-time buyer	30,482	32,898	20,311
	Former owner-occupier	62,324	66,327	40,856
	Non-occupier household buyer	28,814	24,141	11,682

for 23.7%, 19.6%, and 16.0%.

In the new-build market, by contrast, the split is much more even in the pre-policy period and tilts progressively towards first-time buyers thereafter. First-time buyers account for 38.8% of new-build purchases before the policy, 48.2% under the initial regime, and 54.4% under the enhanced regime. Over the same periods, the share of former owner-occupiers falls from 46.3% to 45.3% and then to 41.1%, whilst the share of non-occupier household buyers declines more sharply from 14.9% to 6.4% and then to 4.5%. These totals therefore suggest that first-time buyers became increasingly prominent within the new-build segment over the policy period. Because the three policy regimes cover different numbers of months, these totals reflect cumulative volumes rather than comparable measures of transaction intensity.

6 Empirical Strategy

6.1 Bunching Estimator

To quantify the extent of threshold responses at €500,000, I estimate the counterfactual transaction-value distribution using the bunching framework of Best and Kleven (2018). The bunching estimator recovers the smooth counterfactual transaction-value distribution that would have prevailed in the absence of the HTB notch and compares it with the observed distribution in a neighbourhood of the cutoff.

Let $\bar{v}$ = €500,000 denote the HTB eligibility threshold, and let c_i be the number of transactions in transaction-value bin i, where transaction-values are grouped into €5,000 bins. Let z_i denote the distance between the midpoint of bin i and the final eligible bin [€495,000, €500,000). I assign transactions priced at exactly €500,000 to [€500,000, €505,000), since new-build developers overwhelmingly price just below the threshold rather than at the round-number itself[12]. I estimate the counterfactual distribution using

$$c_i = \sum_{j=0}^{q} \beta_j (z_i)^j + \sum_{r \in R} \eta_r \mathbf{1}\left\{ \frac{\bar{v} + z_i}{r} \in \mathbb{N} \right\} + \sum_{k=h_{\bar{v}}^-}^{h_{\bar{v}}^+} \gamma_k \mathbf{1}\{i = k\} + e_i, \tag{11}$$

where q is the polynomial order, R denotes the set of round-number controls, $\mathbb{N}$ denotes the set of natural numbers, and $\mathbf{1}\{\cdot\}$ is the indicator function. The second term includes fixed effects for focal transaction-values, whilst the third excludes the region $(h_{\bar{v}}^-, h_{\bar{v}}^+)$ around the threshold that may be distorted by bunching responses, where $h_{\bar{v}}^-$ and $h_{\bar{v}}^+$ denote the lower and upper excluded bins respectively.

In the baseline specification, I set $q = 5$, use a bin width of €5,000, exclude one bin on either side of the notch, and set $R = \{10{,}000, 25{,}000\}$[13]. Sensitivity to the polynomial

[12] Among transactions in the €450,000–€550,000 window, 3.4% of new-build transactions (501 sales) are priced in the €499,000–€499,999 range against 0.03% (2 sales) at exactly €500,000. In the second-hand market, 0.3% (49 sales) are priced in the €499,000–€499,999 range against 6.2% (1,074 sales) at exactly €500,000. New-build developers therefore avoid the round-number that second-hand sellers gravitate towards, consistent with active threshold avoidance.

[13] The set of round-number controls is not identical across all specifications. In the second-hand

order, bin width, exclusion region, and number of bootstrap replications is examined in Section 8.

The estimation window is restricted to transaction-values between €300,000 and €600,000, which yields 40 bins below the threshold and 20 above it. The lower bound is chosen so that the polynomial counterfactual is fitted over a region with substantial new-build mass, whilst remaining sufficiently tight that the polynomial reflects the local shape of the distribution near €500,000 rather than the broader recovery in lower-value new-build construction. The upper bound reflects the sparsity of new-build transactions above €600,000, since extending the window further would add noise without contributing useful identifying variation[14].

The counterfactual number of transactions in each bin, $\hat{c}_i$, is obtained by setting the coefficients on the excluded-region dummies to zero. Excess bunching is defined as the difference between observed and counterfactual bin counts in the excluded bins below the threshold:

$$\hat{B} = \sum_{i \in \mathcal{B}^-} (c_i - \hat{c}_i), \tag{12}$$

where $\mathcal{B}^-$ ends with the final eligible bin [€495,000, €500,000). Missing mass is defined analogously over the excluded bins at and above the threshold, beginning with [€500,000, €505,000):

$$\hat{M} = \sum_{i \in \mathcal{B}^+} (\hat{c}_i - c_i). \tag{13}$$

I also report the normalised bunching estimate

$$\hat{b} = \frac{\hat{B}}{\hat{c}_0}, \tag{14}$$

where $\hat{c}_0$ is the estimated counterfactual count in the final eligible bin. This normalisation makes bunching estimates comparable across samples with different baseline transaction volumes.

The baseline specification is estimated separately for six regime-by-property-type cells, combining three policy regimes (pre-policy, initial, and enhanced) with two property types (new-build and second-hand). I report $\hat{B}$, $\hat{M}$, and $\hat{b}$, which are the objects that

counterfactual exercise in Appendix B, I include controls at €10,000, €25,000, and €50,000, to allow more flexibly for focal-value heaping when the counterfactual is estimated directly from the observed second-hand distribution. In the polynomial bunching estimates, by contrast, the estimation routine permits only two round-number controls, so I use €10,000 and €25,000 as a parsimonious adjustment for focal-value clustering near the notch.

[14] The wider €200,000–€600,000 window is used for the distributional difference-in-differences exercise in Section 6.3, where the aim is to characterise the response across the broader transaction-value distribution rather than to fit a smooth counterfactual at a single point.

map directly to the transaction-value responses in my model, rather than a structural elasticity in the sense of Kleven and Waseem (2013). That elasticity is more natural in tax-notch settings with a well-defined marginal net-of-tax rate, whereas in the present setting a reduction in transaction-value may reflect adjustment in housing characteristics as well as in the posted value itself.

An alternative counterfactual strategy that replaces the polynomial benchmark with the second-hand distribution is reported in Appendix B. Inference is based on the bootstrap distribution of the estimator, and I report percentile bootstrap confidence intervals and bootstrap p-values throughout.

6.2 Implied Transaction-Value Response

Let $\widehat{\Delta v^*}$ denote the empirical counterpart of Δv^*. Equation (10) establishes that $\Delta v^* \approx B(\bar{v})/f(\bar{v})$. In the discrete-bin setting, the counterfactual density at the threshold is approximated by $\hat{f}(\bar{v}) = \hat{c}_0/BW$, where $\hat{c}_0$ is the counterfactual count in the threshold bin and BW is the bin width. Substituting into (10) gives

$$\begin{aligned} \widehat{\Delta v^*} &\approx \frac{\hat{B}}{\hat{c}_0/BW} \\ &= \hat{b} \times BW, \end{aligned} \tag{15}$$

where $\hat{b} = \hat{B}/\hat{c}_0$ is the normalised excess mass from (14). This quantity measures the implied transaction-value response of the marginal developer who relocates from above the threshold to at or below it. Percentile bootstrap confidence intervals and p-values are obtained from the distribution of $\hat{b}_m \times BW$ across bootstrap replications. Sensitivity to the choice of bin width is examined in Section 8.2.

6.3 Distributional Difference-in-Differences

The bunching estimator quantifies excess mass at the €500,000 notch, but does not show how that response is distributed across the wider transaction-value distribution. As emphasised by Anagol et al. (2024), bunching responses need not take the form of a spike at the threshold with zero density immediately above it, but may instead be diffuse in the presence of frictions. I therefore complement the bunching estimates with a stacked distributional difference-in-differences specification that estimates bin-by-bin treatment effects across the full €200,000–€600,000 window.

I treat the 1st of January 2017 and the 23rd of July 2020 as separate treatment dates and stack them into a common event-time design. For both cohorts, 2014, 2015, and

2016 correspond to $e = -3$, $e = -2$, and $e = -1$ respectively, with $e = -1$ omitted[15]. For Cohort 1, $e = 0$, $e = 1$, and $e = 2$ correspond to 2017, 2018, and 2019. For Cohort 2, $e = 0$ covers the twelve months following the 23rd of July 2020, and $e = 1$ covers the subsequent twelve months, after which the sample ends.

Transactions are grouped into €5,000 bins in the threshold region [€480,000, €520,000), where any bunching or missing-mass response is most likely to appear, and coarser €50,000 bins elsewhere[16]. For each bin j, I estimate a separate linear probability model of the form

$$\begin{aligned}
\mathbf{1}\{v_i \in j\} = {} & \beta_j^I \big(NB_i \times Initial_{ck}\big) + \beta_j^E \big(NB_i \times Enhanced_{ck}\big) \\
& + \gamma_j^I \big(NB_i \times e_{ck} \times \mathbf{1}\{k = I\}\big) + \gamma_j^E \big(NB_i \times e_{ck} \times \mathbf{1}\{k = E\}\big) \\
& + \delta_j^I \big(NB_i \times \max\{e_{ck}, 0\} \times \mathbf{1}\{k = I\}\big) \\
& + \delta_j^E \big(NB_i \times \max\{e_{ck}, 0\} \times \mathbf{1}\{k = E\}\big) \\
& + \alpha_{cmk} + \varepsilon_{ijcmk}, \qquad (16)
\end{aligned}$$

where $\mathbf{1}\{v_i \in j\}$ is an indicator equal to one if transaction i falls in bin j, NB_i indicates a new-build transaction, $k \in \{I, E\}$ indexes the two treatment cohorts, $Initial_{ck}$ and $Enhanced_{ck}$ are cohort-specific post-treatment indicators, e_{ck} denotes event time relative to the relevant reform date, and α_{cmk} denotes county × year-month × cohort effects. Standard errors are clustered at the county level.

The coefficients β_j^I and β_j^E capture the discrete change at the reform date in the probability that a new-build transaction falls in bin j, relative to second-hand transactions in the same county, month, and cohort. The γ terms allow the new-build versus second-hand difference in each bin to evolve linearly within cohort over event time, whilst the δ terms allow that relative trend to change after treatment. I treat this as the preferred specification, since it identifies the policy effect from a discrete break at the treatment date rather than from broader divergence in the new-build versus second-hand transaction-value distribution over the sample period.

The identifying assumption is that, conditional on these trend terms, the probability that a new-build transaction falls in a given bin would have continued to evolve smoothly relative to second-hand transactions through the reform date in the absence

[15] The year 2013, which would correspond to $e = -4$, is excluded from the stacked design. Unlike the bunching estimator, the distributional difference-in-differences relies on parallel trends between new-build and second-hand transactions over time. Including 2013 would extend the pre-treatment window into a period in which the new-build market was still recovering from the post-crisis collapse in construction, when new-build and second-hand transaction counts evolved very differently.

[16] As in the polynomial bunching estimator, transactions priced at exactly €500,000 are assigned to the [€500,000, €505,000) bin rather than to the final eligible bin [€495,000, €500,000), for the reasons discussed in Section 6.1.

of HTB. To assess this assumption, I estimate event-study coefficients at $e = -3$ and $e = -2$ for each bin and combine their p-values using Fisher's method. Section 8.3 reports the results. Inference is based on heteroskedasticity-robust p-values from county-clustered standard errors.

6.4 Buyer-Participation

To assess whether HTB also altered who purchases new-build dwellings, I estimate two difference-in-differences buyer-participation designs. The first asks whether the first-time buyer share rises more strongly in new builds than in second-hand dwellings. The second asks whether first-time buyers become more concentrated in the new-build segment relative to former owner-occupiers. As in the distributional difference-in-differences design, the initial and enhanced regime indicators are mutually exclusive, so each interaction coefficient is interpreted relative to the pre-policy period.

6.4.1 First-Time Buyer Share: New-Build versus Second-Hand

The first design compares the evolution of the first-time buyer share in new-build and second-hand transactions. Let FTB_{rdt} denote the number of first-time buyer purchases in RPPI region r, dwelling-status cell d, and month t, and let HH_{rdt} denote the total number of household purchases in that cell. I define the first-time buyer share as

$$Share^{FTB}_{rdt} = 100 \times \frac{FTB_{rdt}}{HH_{rdt}}.$$

I estimate

$$\begin{aligned} Share^{FTB}_{rdt} &= \alpha + \beta_1\big(NB_d \times Initial_t\big) + \beta_2\big(NB_d \times Enhanced_t\big) \\ &\quad + \mu_{rd} + \phi_t + u_{rdt}, \end{aligned} \tag{17}$$

where NB_d is an indicator for the new-build segment, $Initial_t$ equals one from the 1st of January 2017 to the 31st of July 2020, and $Enhanced_t$ equals one from the 1st of August 2020 onwards[17]. The omitted category is the pre-policy period, from the 1st of January 2013 to the 31st of December 2016. The region-by-dwelling fixed effects μ_{rd} absorb time-invariant differences between new-build and second-hand markets within each region.

The coefficients of interest are β_1 and β_2. Positive estimates indicate that the first-time buyer share rose more strongly in the new-build segment than in the second-hand

[17] The enhanced regime formally began on the 23rd of July 2020. Because the buyer-participation data is observed at monthly frequency, I code August 2020 as the first month of this policy period.

segment after the introduction of HTB.

The identifying assumption is that, absent HTB, the first-time buyer share of new-build transactions would have evolved in parallel with the corresponding share in second-hand transactions[18]. I assess this assumption in Section 8 using raw annual series, quarterly event-study estimates with the fourth quarter of 2016 as the reference period, and a joint Wald test of the pre-treatment interaction coefficients. Inference is based on p-values from region-clustered standard errors.

6.4.2 New-Build Share: First-Time Buyers versus Former Owner-Occupiers

The second design reverses the comparison. Rather than comparing dwelling types within a buyer group, it compares buyer groups within dwelling types. Let NB_{rbt} and SH_{rbt} denote, respectively, the numbers of new-build and second-hand purchases for buyer group $b \in \{FTB, Former\}$ in region r and month t. I define the new-build share as

$$Share^{NB}_{rbt} = 100 \times \frac{NB_{rbt}}{NB_{rbt} + SH_{rbt}}.$$

I estimate

$$\begin{aligned} Share^{NB}_{rbt} &= \alpha + \delta_1\big(FTB_b \times Initial_t\big) + \delta_2\big(FTB_b \times Enhanced_t\big) \\ &\quad + \mu_r + \phi_t + \theta_{rb}t + v_{rbt}, \end{aligned} \tag{18}$$

where FTB_b indicates the first-time buyer group and former owner-occupiers form the comparison group. The region-by-buyer linear time trends $\theta_{rb}t$ allow the new-build share of each buyer group to evolve differently over time within each region even in the absence of HTB.

The coefficients of interest are δ_1 and δ_2. Positive estimates indicate that first-time buyers moved towards new-build dwellings more strongly than former owner-occupiers after the introduction of HTB.

The identifying assumption is that, absent HTB, the new-build share of first-time buyer transactions would have evolved in parallel with the corresponding share for former owner-occupiers, conditional on these region-by-buyer linear time trends[19]. I

[18] The Central Bank of Ireland's macro-prudential mortgage rules were also revised from the 1st of January 2017, coinciding with the introduction of HTB (Central Bank of Ireland, 2016). The main change simplified the first-time buyer loan-to-value limit from a two-tier structure (90% on the first €220,000, 80% above) to a flat 90% on the full property value. This change applied equally to new-build and second-hand purchases, so it differences out in a design that compares dwelling types within this buyer group. The enhanced HTB regime in July 2020 was not accompanied by any change in mortgage lending rules.

[19] The concurrent 2017 revision of macro-prudential regulations had different impacts on first-time buyers compared with former owner-occupiers. Yet, because this reform applied in the same way to

again assess this assumption in Section 8 using raw annual series, quarterly event-study estimates with the fourth quarter of 2016 as the reference period, and a joint Wald test of the pre-treatment interaction coefficients. Inference is again based on p-values from region-clustered standard errors.

The two designs approach the same underlying question from complementary margins. If HTB draws first-time buyers into the new-build segment, this should appear both as a rising first-time buyer share within new builds and as a rising new-build share within first-time buyer purchases. Agreement across the two designs would therefore strengthen the interpretation that the shift is policy-driven rather than an artefact of the particular comparison group or dependent variable chosen.

7 Results

This section reports the empirical evidence on responses to the €500,000 HTB threshold. I first examine threshold bunching and the implied transaction-value response, then turn to distributional evidence on where the response appears in transaction-value space, and finally examine whether buyer-participation in the new-build segment moved towards first-time buyers after the introduction of HTB.

7.1 Bunching Estimator

The bunching estimates provide the first direct test of prediction (i) of the model, namely excess mass at or just below the €500,000 notch in the treated new-build market.

Figure 6 plots the observed bin counts against the fitted counterfactual over the €300,000–€600,000 window for each regime-by-market cell. Visually, the treated new-build panels show excess mass in the final eligible bin and missing mass at and above the threshold, whereas the pre-policy and second-hand panels do not display a comparable pattern beyond ordinary round-number heaping that is present elsewhere in the distribution.

Table 4 formalises this comparison. Under both policy regimes, the treated new-build estimates are positive and statistically significant at the 10% level, though imprecisely estimated. This reflects the scale of the Irish housing market, and the new-build segment in particular, which is considerably smaller than settings such as the United Kingdom stamp-duty market studied by Best and Kleven (2018). The normalised

both new-build and existing properties within each buyer category, it does not generate a relative increase in new-build purchases specifically among first-time buyers.

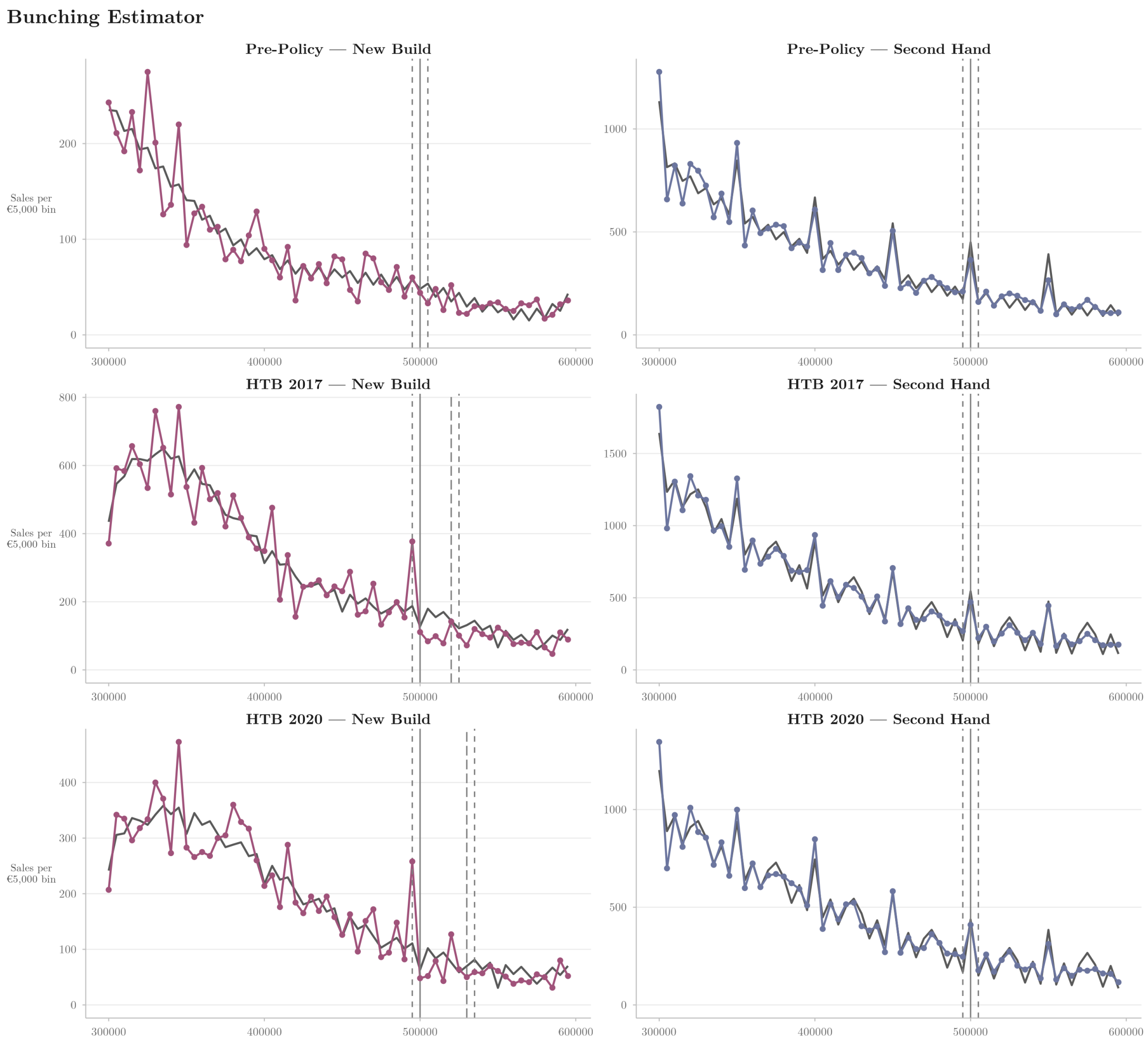

Figure 6: Observed and counterfactual transaction-value distributions around the €500,000 notch, with round-number controls.

Table 4: Bunching Estimates

	Pre-Policy	HTB 2017	HTB 2020
Panel A: New Build			
$\hat{b}$	-0.417	1.352*	2.090*
	[-2.020, 1.053]	[-0.229, 4.685]	[-0.231, 9.343]
$\hat{B}$	-22.1	172.7	132.1
	[-107.2, 55.9]	[-29.3, 598.4]	[-14.6, 590.4]
$\hat{M}$	14.5	407.1	162.7
$\hat{B}/\hat{M}$	-1.53	0.42	0.81
Panel B: Second Hand			
$\hat{b}$	-0.177	0.091	0.307
	[-0.873, 0.626]	[-0.624, 0.886]	[-0.397, 1.176]
$\hat{B}$	-46.2	28.3	77.4
	[-227.3, 163.0]	[-193.8, 275.2]	[-100.1, 296.6]
$\hat{M}$	-61.8	-10.5	17.3
$\hat{B}/\hat{M}$	0.75	-2.70	4.47
Polynomial	5	5	5
RN controls	Yes	Yes	Yes

Notes: Polynomial counterfactual from Eq. (11) with round-number controls at €10,000 and €25,000. 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

bunching coefficient is $\hat{b} = 1.352$ under the initial regime and 2.090 under the enhanced regime. By contrast, the pre-policy new-build estimate is $\hat{b} = -0.417$ and statistically indistinguishable from zero, whilst the second-hand estimates are small, unstable in sign, and statistically insignificant in every period. The threshold response therefore appears only where the policy changes the developer's problem, which is in line with prediction (i). Before HTB and in the second-hand market, where $E(\tau, \lambda) = 0$ by construction, pricing above $\bar{v}$ does not generate the same discrete loss in effective demand.

The larger estimate under the enhanced regime is in the direction predicted by the comparative static on τ in Section 4: a larger τ increases the demand loss from pricing above $\bar{v}$, and therefore strengthens the incentive to bunch below the cutoff. The increase in $\hat{b}$ from 1.352 to 2.090, or approximately 55%, is in line with the 50% increase in the maximum subsidy at the threshold from €20,000 to €30,000, consistent with a local response of bunching intensity that strengthens with the value of eligibility lost above the cutoff. This is consistent with prediction (ii), and with the finding in Uribe (2022) that bunching at a housing-subsidy threshold increases monotonically with subsidy generosity. Because $\hat{b}$ scales excess mass by the counterfactual count in the threshold bin, it is the relevant object for comparing bunching intensity across regimes with different sample lengths and transaction volumes. The corresponding excess-mass estimates are $\hat{B} = 172.7$ under the initial regime and 132.1 under the enhanced regime. The lower level of $\hat{B}$ under the enhanced regime reflects the shorter sample window rather than a weaker behavioural response.

The relationship between excess mass below the threshold and missing mass above it is informative about the form of adjustment. In the frictionless benchmark, mass withdrawn from just above $\bar{v}$ reappears at the cutoff, so one would expect a close correspondence between excess bunching below the threshold and missing mass above it. Empirically, however, the polynomial estimator gives $\hat{B}/\hat{M} = 0.42$ under the initial regime and 0.81 under the enhanced regime, so missing mass exceeds excess mass in both cases. This gap is consistent with two non-exclusive channels. On the intensive margin, developers may adjust the unit price p or housing characteristics h across a range above the threshold, so that the response is spread over multiple bins rather than concentrated in the final eligible bin. On the extensive margin, some units with latent transaction-values just above the threshold may not be developed at that specification, with developers redirecting resources towards units at other price points, though this channel cannot be separately identified.

The extensive-margin channel of foregone above-threshold units, taken alone, would sit awkwardly with the buyer-participation evidence in Section 7.4, which shows that

HTB draws additional first-time buyers into the new-build segment. The two are consistent if the segment-level participation response is concentrated in a different part of the transaction-value distribution from the threshold response, and the distributional difference-in-differences in Section 7.3 supports this reading: the relative new-build distribution rises in the middle of the eligible range (€350,000–€450,000) and falls in the lower range (€200,000–€250,000), so incoming first-time buyers appear to transact below rather than around the cutoff.

The increase in $\hat{B}/\hat{M}$ from the initial to the enhanced regime suggests that the adjustment nonetheless becomes more concentrated near the cutoff as subsidy generosity rises, in line with the model's prediction that a larger τ intensifies the threshold distortion. Relative to Agarwal, Hu, and Lee (2023), where excess mass exceeds missing mass and is interpreted as evidence of net entry, this pattern of $\hat{B}/\hat{M} < 1$ instead suggests that not all missing mass above the threshold is recovered below it, whether because the response is spread across multiple adjustment margins or because some above-threshold activity is foregone entirely.

Under the initial regime, the normalised bunching coefficient of 1.352 is somewhat below the range of 1.64–1.85 reported by Best and Kleven (2018) at stamp-duty notches in the United Kingdom, and substantially smaller than the 8.39 reported by Agarwal, Hu, and Lee (2023) for New South Wales, Australia. Under the enhanced regime, the estimate of 2.090 exceeds Best and Kleven (2018)'s range. This cross-regime pattern is consistent with the structure of HTB. Only a fraction of demand is exposed to the notch ($\lambda < 1$), which attenuates the response relative to universal-notch settings, but a sufficiently large rise in τ under the enhanced regime may be sufficient to outweigh this attenuation.

The sensitivity of these estimates to the polynomial order, bin width, exclusion zone, and the number of bootstrap replications is examined in Section 8, and an alternative counterfactual using the second-hand distribution directly is reported in Appendix B.

7.2 Implied Transaction-Value Response

The bunching estimates establish that HTB generates excess mass below the €500,000 cutoff. The next step is to translate the response into the transaction-value object of the model. Table 5 therefore converts the bunching estimates into implied transaction-value responses at the notch using Equation (10).

The implied response of the marginal buncher is $\widehat{\Delta v^*} =$ €6,759 under the initial regime and €10,448 under the enhanced regime, corresponding to 33.8% and 34.8% of the maximum statutory relief τ in each regime.

Table 5: Implied Transaction-Value Response

	HTB 2017	HTB 2020
$\hat{b}$	1.352*	2.090*
	[-0.229, 4.685]	[-0.231, 9.343]
$\widehat{\Delta \bar{v}}$	€6,759	€10,448
	[€-1,147, €23,427]	[€-1,156, €46,713]
$\widehat{\Delta \bar{v}}/\tau$	33.8%	34.8%
	[-5.7%, 117.1%]	[-3.9%, 155.7%]
τ	€20,000	€30,000
Bin width	€5,000	€5,000

Notes: 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

These magnitudes are consistent with prediction (iii) of the model. The marginal buncher's indifference condition equates the profit gain from preserving eligibility with the profit loss from lowering v, so the implied transaction-value adjustment need not approach the full subsidy. Because $v = p \cdot h$ bundles the unit price with housing characteristics, the response reflects adjustment in the transacted bundle rather than a pure reduction in the unit price. The ratio of $\widehat{\Delta v^*}$ to τ is around one-third in both regimes, in line with a marginal-buncher response that strengthens with the value of eligibility, consistent with the comparative static on τ. The estimates remain smaller than the two-to-five multiples of the tax increase reported by Best and Kleven (2018) at stamp-duty notches in the United Kingdom, which is again consistent with the more limited scope of HTB.

The sensitivity of the implied response to the choice of bin width is examined in Section 8.2, and estimates under the second-hand counterfactual are reported in Appendix B.

7.3 Distributional Difference-in-Differences

This subsection locates the €500,000 threshold distortion within the wider transaction-value distribution and assesses how tightly it is concentrated around the notch. I focus on Column (2), the preferred specification, which includes cohort-specific linear trends and post-treatment trend breaks for new-build transactions.

Table 6 reports the stacked distributional difference-in-differences estimates, whilst Figure 7 summarises the same pattern graphically. Figure 7 highlights two features of the estimates. The left-hand panel shows a broader upward movement in the new-build transaction-value distribution relative to second hand, whereas the right-hand

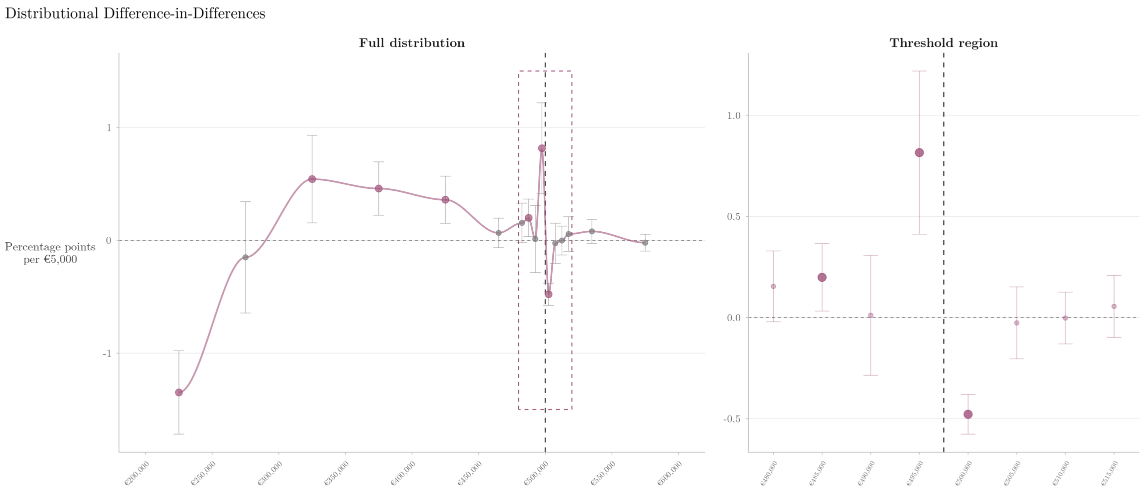

Figure 7: Pooled estimates normalised to percentage points per €5,000. The left panel reports the full distribution and the right panel shows the threshold region.

panel isolates the clear localised break at $\bar{v}$ by focusing on the threshold region.

Around the threshold, the estimates in Table 6 line up closely with the bunching results. In the last eligible bin, [€495,000, €500,000), the coefficient is 0.476 under the initial regime and 1.181 under the enhanced regime, equivalent to increases of 79.3% and 196.8% relative to the mean bin share of 0.60%. These estimates are significant at the 1% and 5% levels respectively. In the first ineligible bin, [€500,000, €505,000), the coefficient is −0.423 under the initial regime and −0.538 under the enhanced regime, or 65.1% and 82.8% of the mean share of 0.65%, with both estimates significant at the 1% level. The local sign pattern therefore matches prediction (i): mass rises immediately below the cutoff and falls immediately above it. The stronger below-threshold response under the enhanced regime is likewise consistent with prediction (ii), namely stronger responses under the more generous schedule.

The bin-level estimates also show that the response is highly local. Above the threshold, it is concentrated almost entirely in the first ineligible bin: the coefficients on [€505,000, €510,000), [€510,000, €515,000), and [€515,000, €520,000) are all small and statistically insignificant in both regimes. Below the threshold, the response is somewhat less concentrated. Under the enhanced regime, the coefficient in [€485,000, €490,000) is 0.333, significant at the 5% level and equal to 61.7% of the mean share in that bin, whereas the coefficient on [€490,000, €495,000) is indistinguishable from zero. The [€485,000, €490,000) coefficient should be interpreted with some caution, since it is the only bin in the immediate neighbourhood of the cutoff for which the joint Fisher p-value reported in Table 10 rejects the null of no differential pre-trend. The pattern is therefore one of a sharply local distortion with limited spillover into

Table 6: Distributional Difference-in-Differences

	Mean	(1) Initial	(1) Enhanced	(2) Initial	(2) Enhanced
[€200,000, €250,000)	25.81%	-14.549***	-15.341***	-11.398***	-15.729***
		(1.716)	(2.071)	(1.647)	(2.261)
[€250,000, €300,000)	22.97%	-3.128	-3.349	0.876	-4.076
		(2.647)	(3.476)	(2.194)	(3.521)
[€300,000, €350,000)	17.32%	5.999***	4.149	5.726***	5.102
		(1.838)	(3.619)	(1.536)	(3.590)
[€350,000, €400,000)	12.37%	6.425***	7.189***	3.000**	6.298***
		(1.344)	(2.319)	(1.220)	(1.843)
[€400,000, €450,000)	8.34%	2.889***	4.298**	2.002**	5.311**
		(0.825)	(2.078)	(0.792)	(2.022)
[€450,000, €480,000)	3.84%	1.186***	1.134*	-0.244	1.075*
		(0.411)	(0.627)	(0.265)	(0.575)
[€480,000, €485,000)	0.55%	0.086	-0.004	0.181	0.126
		(0.078)	(0.144)	(0.160)	(0.224)
[€485,000, €490,000)	0.54%	0.273**	0.470**	0.074	0.333**
		(0.104)	(0.213)	(0.091)	(0.154)
[€490,000, €495,000)	0.47%	0.153***	0.075	0.019	0.003
		(0.053)	(0.126)	(0.173)	(0.159)
[€495,000, €500,000)	0.60%	**1.035*** **	**1.282*** **	**0.476*** **	**1.181** **
		(0.299)	(0.447)	(0.088)	(0.431)
[€500,000, €505,000)	0.65%	**-0.262*** **	**-0.444*** **	**-0.423*** **	**-0.538*** **
		(0.033)	(0.048)	(0.092)	(0.034)
[€505,000, €510,000)	0.32%	-0.029	0.014	-0.052	0.002
		(0.084)	(0.125)	(0.117)	(0.090)
[€510,000, €515,000)	0.44%	-0.060	0.045	0.006	-0.012
		(0.040)	(0.099)	(0.062)	(0.127)
[€515,000, €520,000)	0.29%	0.013	-0.022	-0.001	0.117
		(0.033)	(0.079)	(0.048)	(0.117)
[€520,000, €550,000)	2.12%	0.118	0.660	0.070	0.911
		(0.145)	(0.484)	(0.289)	(0.812)
[€550,000, €600,000)	3.37%	-0.148	-0.157	-0.312	-0.102
		(0.200)	(0.713)	(0.268)	(0.638)
Observations		256,681			
County × year-month × cohort FE		Yes	Yes	Yes	Yes
NB × cohort-specific trend		No	No	Yes	Yes
NB × cohort-specific trend break		No	No	Yes	Yes
Clustered SEs (county)		Yes	Yes	Yes	Yes

Notes: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

the adjacent lower eligible range, consistent with Best and Kleven (2018)'s observation that optimisation frictions may be less important in housing markets than in labour markets.

Away from the cutoff, the estimates point to broader relative movements in the new-build distribution. In the lower part of the distribution, the coefficient in [€200,000, €250,000) is strongly negative in both regimes. In the middle of the distribution, the coefficients in [€350,000, €400,000) and [€400,000, €450,000) are positive and sizeable. These movements are less easily interpreted through the notch mechanism at $\bar{v}$. They may reflect incentives elsewhere in the HTB schedule, particularly the kink at €400,000 under the initial regime and the kink at €300,000 under the enhanced regime, or the broader upward movement in new-build transaction-values relative to second-hand documented in the summary statistics.

Accordingly, the coefficients away from the threshold should be interpreted more cautiously than those in its immediate neighbourhood. As Cengiz et al. (2019) emphasise in a related bin-level difference-in-differences setting, the credibility of this type of design is strongest in the immediate policy region, where the mechanism is sharpest, rather than in bins further away where broader confounding trends may be more important. The causal interpretation of these estimates therefore rests on the parallel-trends conditions examined in Section 8.3, which are most persuasive in the immediate neighbourhood of the cutoff.

7.4 Buyer-Participation

This subsection evaluates prediction (iv) of the model, namely whether HTB increased the relative importance of first-time buyers within the new-build market. In the model, the threshold distortion in transaction-values depends not only on the value of eligibility, τ, but also on λ, the effective share of demand that is sensitive to the loss of eligibility above the cutoff.

7.4.1 First-Time Buyer Share: New-Build versus Second-Hand

Table 7 reports estimates of Equation (17), which compares the evolution of the first-time buyer share in new-build and second-hand purchases. I focus on Column (2), the preferred specification, because the region-by-dwelling fixed effects absorb time-invariant differences between the new-build and second-hand segments within each region. Column (1) yields very similar estimates, so the qualitative conclusion is unchanged.

In the preferred specification, the coefficients on $NB_d \times Initial_t$ and $NB_d \times Enhanced_t$

Table 7: First-time buyer share of transactions: new build vs second hand

	First-time buyer share (%)	
	(1)	(2)
New Build	13.32***	
	(1.10)	
New Build × HTB 2017	5.01***	5.11***
	(1.54)	(1.55)
New Build × HTB 2020	10.55***	10.50***
	(1.93)	(1.89)
Dep. var. mean	33.33	33.33
Observations	7,322	7,322
Adj. R^2	0.291	0.311
Adj. within R^2	0.210	0.013
Region FE	Yes	
Region × Dwelling FE		Yes
Month FE	Yes	Yes
Clustered SEs	Region	Region
No. regions	33	33

Notes: Each regime coefficient is the full effect relative to the pre-policy period. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

are 5.11 and 10.50 percentage points respectively, and both are statistically significant at the 1% level. These estimates imply that, relative to the pre-policy period, the increase in the first-time buyer share was 5.11 percentage points larger in new builds than in second-hand properties under the initial regime, and 10.50 percentage points larger under the enhanced regime. Relative to the dependent-variable mean of 33.33%, these effects amount to 15.3% and 31.5% of the mean respectively. The enhanced-regime estimate is therefore 2.05 times as large as the initial-regime estimate, in line with the doubling of the marginal subsidy rate from 5% to 10%. The identifying assumptions for this design are examined in Section 8, and broadly support a causal interpretation of the estimated effects.

7.4.2 New-Build Share: First-Time Buyers versus Former Owner-Occupiers

Table 8 reports estimates of Equation (18), which compares the evolution of the new-build share for first-time buyers and former owner-occupiers. Again, I focus on Column (2), the preferred specification, because the region-by-buyer linear trends allow the new-build share of first-time buyers and former owner-occupiers to evolve differently over time within each region in the absence of HTB. Column (1) is less flexible, but

Table 8: New build share of transactions: first-time buyers vs former owner-occupiers

	New build share (%)	
	(1)	(2)
First-time buyer	7.05***	7.23***
	(0.67)	(0.96)
First-time buyer × HTB 2017	2.14*	2.48***
	(1.09)	(0.90)
First-time buyer × HTB 2020	4.74***	5.33***
	(1.22)	(1.43)
Dep. var. mean	15.40	15.40
Observations	7,523	7,523
Adj. R^2	0.367	0.495
Adj. within R^2	0.145	0.318
Region FE	Yes	Yes
Month FE	Yes	Yes
Region × Buyer trend		Yes
Clustered SEs	Region	Region
No. regions	33	33

Notes: Each regime coefficient is the full effect relative to the pre-policy period. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

yields the same qualitative conclusion.

In the preferred specification, the coefficients on $FTB_b \times Initial_t$ and $FTB_b \times Enhanced_t$ are 2.48 and 5.33 percentage points respectively, and both are statistically significant at the 1% level. They imply that, relative to the pre-policy period, the increase in the new-build share was 2.48 percentage points larger for first-time buyers than for former owner-occupiers under the initial regime, and 5.33 percentage points larger under the enhanced regime. Relative to the dependent-variable mean of 15.40%, these effects amount to 16.1% and 34.6% of the mean respectively. The enhanced-regime estimate is therefore 2.15 times as large as the initial-regime estimate. The identifying assumptions for this design are examined in Section 8, and are again generally supportive of a causal interpretation of these estimates.

Both designs therefore reach the same conclusion through different margins, with the buyer-participation response scaling by a factor of approximately two across both (2.05 and 2.15). Since most new-build transactions fall in the lower part of the eligible range, where the marginal subsidy rate doubles, this approximate doubling is more closely aligned with the rate change that applies to the bulk of transactions than with the 50% cap increase that applies near the €500,000 threshold. In terms of the model, a more generous subsidy increases λ, the effective share of demand in the new-build segment that is responsive to HTB eligibility, and near the threshold also raises τ. The enhanced regime therefore strengthens the threshold distortion through both channels highlighted in Section 4.

8 Robustness

This section examines the robustness of the main empirical results and the identifying assumptions underlying the difference-in-differences designs.

8.1 Bunching Estimator

The baseline bunching specification fixes the estimation window at €300,000–€600,000, the polynomial order at $p = 5$, the bin width at €5,000, the exclusion zone at one bin on each side of the notch, and 1,000 bootstrap replications. This subsection varies each choice in turn, with exact estimates reported in Appendix Tables 12–15.

8.1.1 Polynomial Order

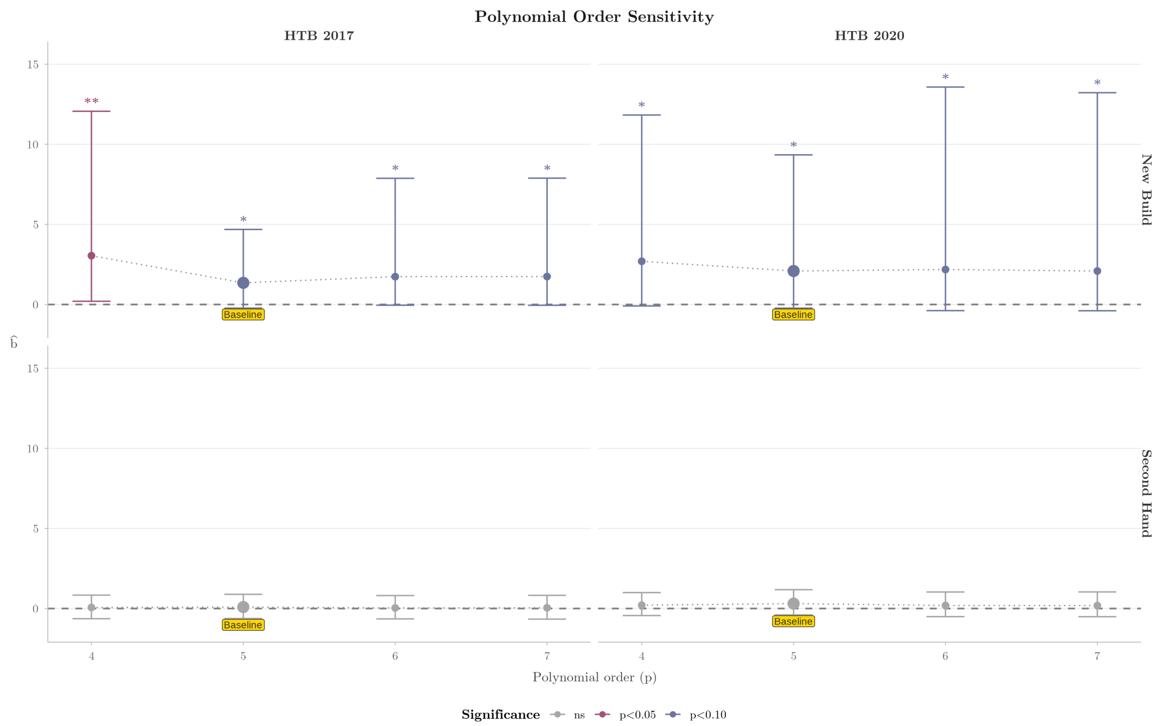

Figure 8: Sensitivity of estimated excess bunching to the choice of polynomial order ($p = 4$ to $p = 7$), shown separately by policy regime and market segment.

Figure 8 shows that the bunching estimates are qualitatively stable across alternative polynomial orders. In the treated new-build sample, the estimated bunching coefficient is positive and statistically significant at every polynomial order considered under both policy regimes, with the $p = 4$ specification under the initial regime significant at the 5% level and the baseline and higher-order specifications significant at 10%. Point estimates are somewhat larger at $p = 4$ than at $p = 5$, but stabilise from $p = 5$ onwards, indicating that the baseline order is not at the edge of the stable region. The second-hand estimates remain close to zero and statistically insignificant throughout.

8.1.2 Bin Width

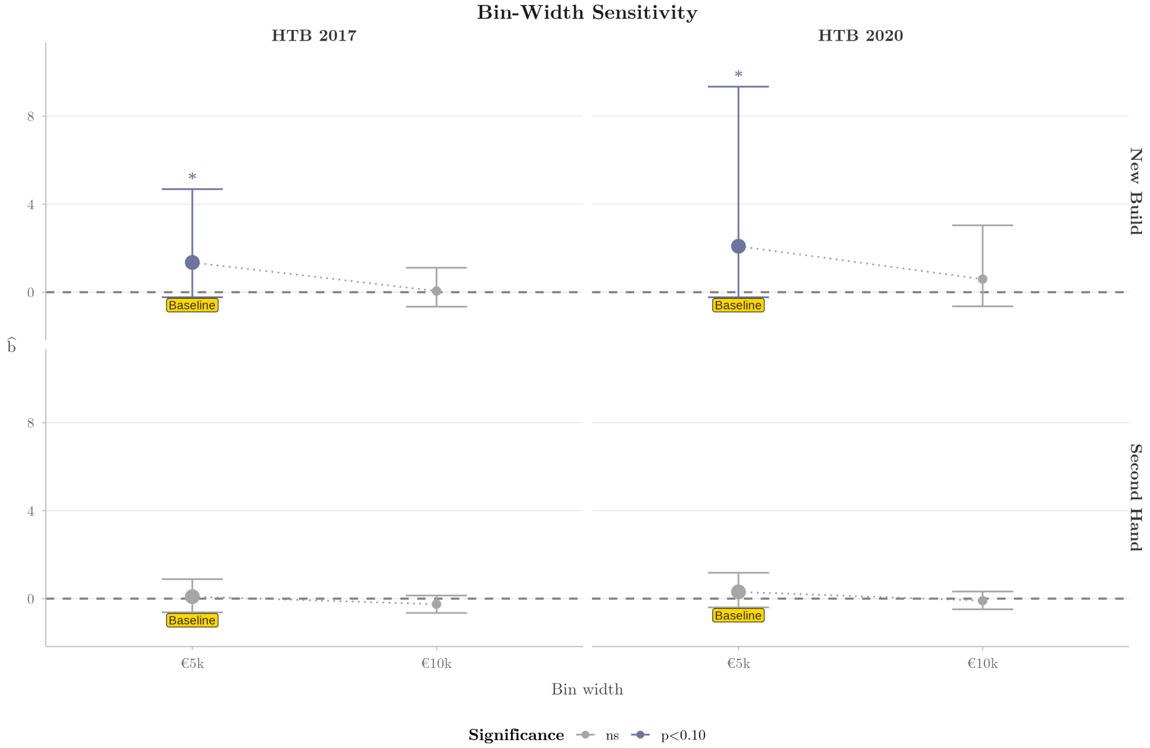

Figure 9: Sensitivity of estimated excess bunching to the choice of bin width (€5,000 and €10,000), shown separately by policy regime and market segment.

Figure 9 shows that the bunching estimates are sensitive to the choice of bin width. In the treated new-build sample, the baseline €5,000 specification produces positive and significant estimates in both policy regimes, whereas the €10,000 specification yields smaller and statistically insignificant point estimates[20]. Two independent sources of evidence account for this pattern. The micro-level pricing data noted in Section 6.1 show that developers cluster in the €499,000–€499,999 range, and the distributional difference-in-differences in Section 7.3 localises the response within

[20]The bin width must divide evenly into the €500,000 threshold for it to fall on a bin boundary. Widths narrower than €5,000 (for example, €2,500) place the threshold in a bin containing too few transactions for a stable counterfactual. Widths coarser than €10,000 yield too few bins to the right of the threshold for the fifth-order polynomial and the two-pass notch-width procedure to converge. This leaves €5,000 and €10,000 as the only feasible specifications.

[€495,000, €500,000), with no detectable spillover into adjacent bins. The threshold response therefore operates at a scale narrower than €5,000, and a €10,000 bin width pools this region with transactions that are unaffected by HTB, mechanically attenuating the estimate. The €5,000 specification is the coarsest feasible resolution at which the distortion can be recovered cleanly. Second-hand estimates show no comparable sensitivity, sitting close to zero under both bin widths.

8.1.3 Exclusion Zone

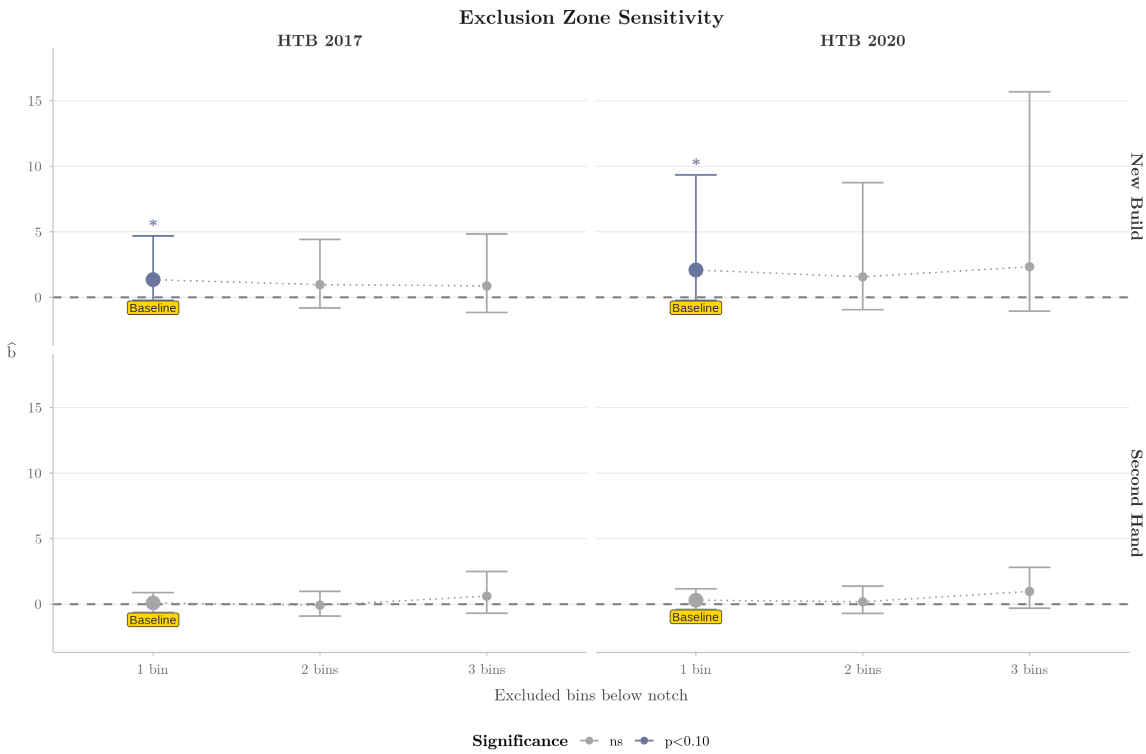

Figure 10: Sensitivity of estimated excess bunching to the width of the exclusion zone around the €500,000 notch, shown separately by policy regime and market segment.

As emphasised by Kleven (2016), sensitivity to the excluded range provides a useful diagnostic for bunching estimates. Figure 10 compares specifications that exclude one, two, and three bins on each side of the threshold. Point estimates in the treated new-build sample remain positive across all three specifications, though precision weakens substantially as the excluded region widens. Under the initial regime, $\hat{b}$ declines slightly from 1.352 at one excluded bin to 0.969 at two and 0.873 at three, and loses significance once more than one bin is excluded. Under the enhanced regime, the estimate declines at two excluded bins and then rebounds at three, but with a confidence interval wide enough that the response is indistinguishable from zero at both. This pattern reflects the narrow scale at which developers cluster below the notch: widening the exclusion zone removes bins that lie close enough to the cutoff to be informative about the threshold response, rather than absorbing additional bunching mass. The baseline specification with a single excluded bin on each side therefore corresponds to the natural scale of the response rather than an arbitrary choice. In the second-hand sample, by contrast, the estimates are uniformly small and indistinguishable from zero regardless of the excluded range.

8.1.4 Bootstrap Replications

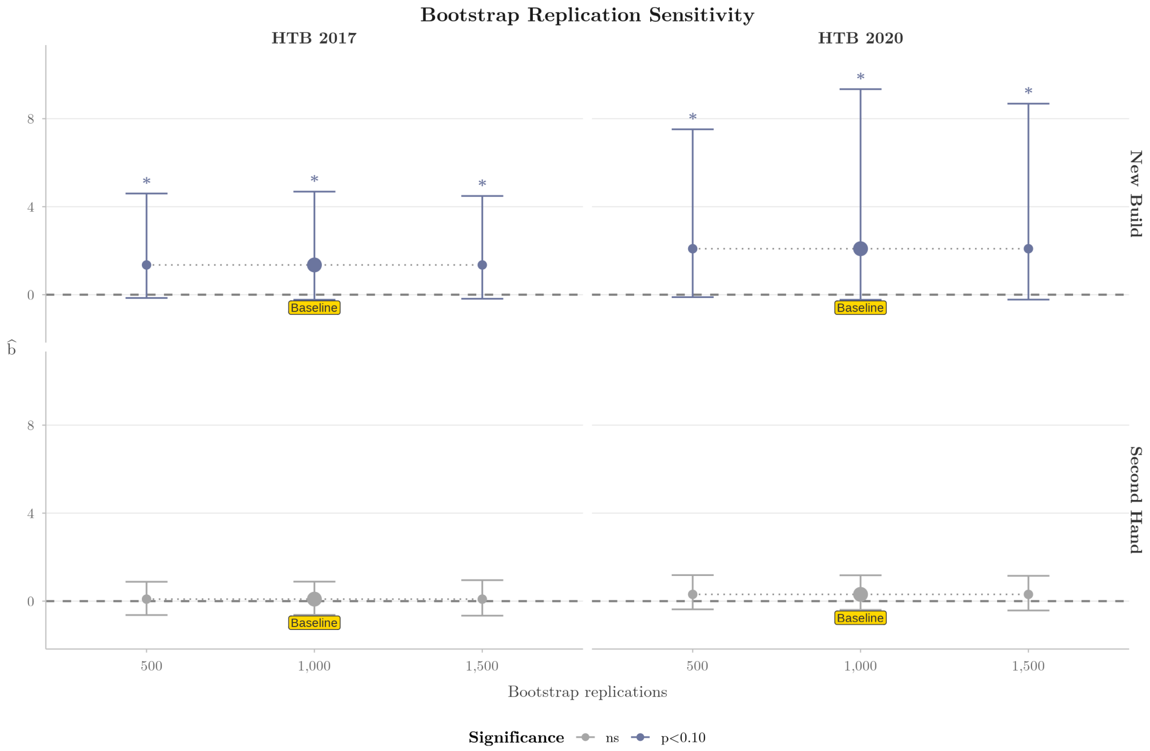

Figure 11: Sensitivity of estimated excess bunching to the number of bootstrap replications (500, 1,000, and 1,500), shown separately by policy regime and market segment.

Figure 11 shows that the bunching point estimates are invariant to the number of bootstrap replications, as expected, since bootstrap resampling affects inference but not the underlying estimate. Confidence intervals and significance are essentially unchanged from 500 to 1,500 replications in every cell, indicating that the baseline choice of 1,000 replications is sufficient to stabilise inference. Second-hand inference is similarly stable, with estimates close to zero across all three replication counts.

Overall, the bunching robustness checks support the baseline specification against variation in polynomial order and bootstrap replications. Across alternative polynomial orders, the treated new-build estimates remain positive and significant under both policy regimes, and bootstrap replications leave inference unchanged. The exclusion zone and bin width exercises show that significance weakens when the specification is moved away from the narrow scale at which developers cluster below the notch, which is itself consistent with the micro-level evidence that the threshold response operates at a scale of approximately €1,000 below the cutoff rather than across a broader range. Second-hand estimates display no comparable response in any specification. An alternative counterfactual using the second-hand distribution directly is examined in Appendix B.

8.2 Implied Transaction-Value Response

Table 9: Implied Transaction-Value Response: Bin-Width Sensitivity

Regime	Bin width	$\hat{b}$	$\widehat{\Delta \bar{v}}$	$\widehat{\Delta \bar{v}}/\tau$
HTB 2017	€5,000	1.352*	€6,759	33.8%
			[€-1,147, €23,427]	[-5.7%, 117.1%]
	€10,000	0.053	€530	2.7%
			[€-6,578, €11,131]	[-32.9%, 55.7%]
HTB 2020	€5,000	2.090*	€10,448	34.8%
			[€-1,156, €46,713]	[-3.9%, 155.7%]
	€10,000	0.594	€5,938	19.8%
			[€-6,414, €30,402]	[-21.4%, 101.3%]

Notes: 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 9 examines the sensitivity of the implied transaction-value response to the choice of bin width. The estimates remain positive under both regimes at both bin widths, and the enhanced-regime estimate exceeds the initial-regime estimate throughout. At the baseline €5,000 bin width, $\widehat{\Delta v^*}$ = €6,759 under the initial regime and €10,448 under the enhanced regime, or 33.8% and 34.8% of the maximum statutory relief. At a €10,000 bin width, the estimates fall to €530 and €5,938 respectively, and neither is statistically distinguishable from zero. This pattern mirrors the bin-width sensitivity of the underlying bunching coefficient reported in Section 8 and reflects the same mechanism.

8.3 Distributional Difference-in-Differences

The distributional difference-in-differences identifies the policy effect from the change in new-build versus second-hand bin shares at the introduction of both phases of HTB, conditional on the cohort-specific trend and trend-break terms in equation (16). The corresponding identifying assumption is that, absent HTB, the new-build versus second-hand difference in each bin would have evolved smoothly through the policy periods. To examine this assumption, I estimate a separate event-study specification for each bin j,

$$\mathbf{1}\{v_i \in j\} = \sum_{e \neq -1} \pi_e^j \big(NB_i \times \mathbf{1}\{e_{ck} = e\}\big) + \alpha_{cmk} + \varepsilon_{ijcmk}, \tag{19}$$

where $e \in \{-3, -2, 0, 1, 2\}$ with $e = -1$ omitted, i indexes transactions, j bins, c counties, m year-months, and k cohorts. The term α_{cmk} denotes county $\times$ year-month $\times$ cohort fixed effects, and standard errors are clustered at the county level. For each bin, I then combine the two pre-treatment p-values using Fisher's method[21] and report the resulting joint p-value in Table 10. The null hypothesis is that the two pre-treatment lead coefficients are jointly equal to zero, implying no differential pre-trend in that bin.

Table 10 indicates that the pre-trends evidence is mixed across the full distribution. Several bins reject the null of no differential pre-trend, particularly in the lower and middle parts of the transaction-value distribution. This is consistent with the descriptive evidence that new-build and second-hand markets were already evolving differently in broader ways before HTB. It also motivates the preferred specification in equation (16), which allows for cohort-specific linear trends and cohort-specific post-treatment trend breaks for new-build transactions rather than imposing common distributional evolution across the two market segments.

In the immediate neighbourhood of the policy cutoff, the pre-trends evidence is more supportive. The joint Fisher p-value does not reject the null in the final eligible bin, the first ineligible bin, or the immediately adjacent bins on either side. The only exception in this local region is [€485,000, €490,000), which rejects at conventional levels and should therefore be interpreted more cautiously.

The implication is that the distributional difference-in-differences is most credible in the narrow region around the cutoff rather than across the full transaction-value distribution. A separate rejection in the upper tail reinforces this point, but does not alter the local interpretation. Across the full distribution, the design is best read as evidence on how new-build transactions evolve relative to second-hand once broader divergence is absorbed by the trend-break specification. Around the cutoff itself, where the pre-trends evidence is generally supportive and the sign pattern aligns closely with the theory, the estimates are more naturally interpreted as causal evidence on the local shape of the HTB threshold response.

[21] Under the null, the statistic $-2\sum_{e<0} \log p_e^j$ is distributed as χ_4^2.

Table 10: Pre-Trends: Stacked Distributional Event Study

	$t = -3$	$t = -2$	Fisher p
[€200,000, €250,000)	-0.32	-9.34***	0.004
	(2.02)	(2.37)	
[€250,000, €300,000)	-0.78	-0.40	0.681
	(0.91)	(1.54)	
[€300,000, €350,000)	6.37***	7.59**	0.001
	(1.91)	(3.48)	
[€350,000, €400,000)	-1.11***	1.26	0.020
	(0.38)	(1.60)	
[€400,000, €450,000)	-1.52**	0.21	0.116
	(0.68)	(0.58)	
[€450,000, €480,000)	-1.72***	0.38	0.006
	(0.52)	(0.33)	
[€480,000, €485,000)	0.18	-0.12	0.151
	(0.15)	(0.08)	
[€485,000, €490,000)	0.29***	-0.00	0.010
	(0.08)	(0.13)	
[€490,000, €495,000)	-0.04	-0.06	0.479
	(0.05)	(0.06)	
[€495,000, €500,000)	**-0.13**	**0.42**	0.148
	(0.11)	(0.27)	
[€500,000, €505,000)	**-0.13**	**0.02**	0.802
	(0.20)	(0.11)	
[€505,000, €510,000)	-0.14	-0.04	0.385
	(0.12)	(0.06)	
[€510,000, €515,000)	-0.30	0.10	0.199
	(0.19)	(0.12)	
[€515,000, €520,000)	-0.07	0.04	0.551
	(0.06)	(0.10)	
[€520,000, €550,000)	-0.87***	0.19	0.015
	(0.30)	(0.16)	
[€550,000, €600,000)	0.28	-0.24	0.373
	(0.30)	(0.23)	
Observations		256,681	
County × month × cohort FE		Yes	
Clustered SEs (county)		Yes	

Notes: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

8.4 Buyer-Participation

This subsection assesses the identifying assumptions underlying the two buyer-participation difference-in-differences designs. In each case, I first inspect the raw annual series and then turn to a quarterly event-study specification, which provides a more formal test of whether the relevant treatment and comparison groups were already diverging before HTB was introduced.

8.4.1 First-Time Buyer Share: New-Build versus Second-Hand

I begin with the design that compares the first-time buyer share in new-build and second-hand transactions. The identifying assumption is that, absent HTB, these two series would have continued to evolve in parallel. Figure 12 provides a first diagnostic by plotting the raw annual series for the two groups.

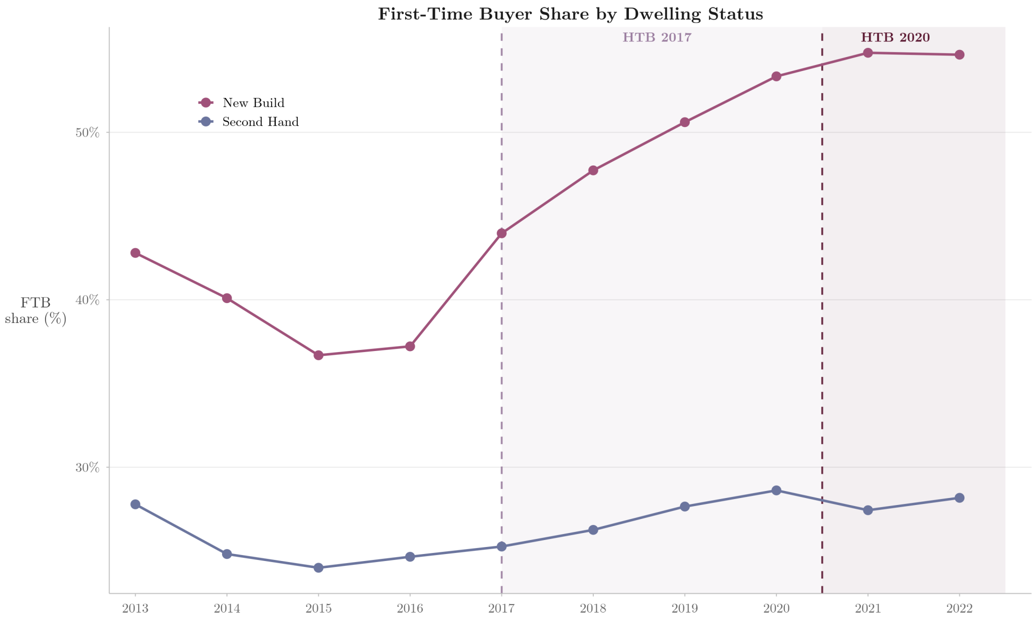

Figure 12: First-time buyer share of transactions by dwelling status, annual.

Figure 12 plots the first-time buyer share separately for new-build and second-hand dwellings. In the pre-policy period, the two series move broadly in parallel, with both shares trending downwards. After the introduction of HTB in 2017, the new-build series rises markedly, whereas the second-hand series increases only modestly.

To examine pre-treatment dynamics more formally, I estimate the following quarterly event-study version of the preferred difference-in-differences specification:

$$Share_{rdt}^{FTB} = \mu_{rd} + \sum_{q \neq 2016Q4} \alpha_q \mathbf{1}\{t \in q\} + \sum_{q \neq 2016Q4} \beta_q \big(NB_d \times \mathbf{1}\{t \in q\}\big) + u_{rdt}, \quad (20)$$

where μ_{rd} denotes a fixed effect for each region $\times$ dwelling cell and standard errors are clustered by RPPI region. The coefficients β_q trace the evolution of the first-time-buyer-share differential between new-build and second-hand transactions over time. The corresponding Wald test evaluates the joint null that all pre-2017 interaction coefficients are equal to zero.

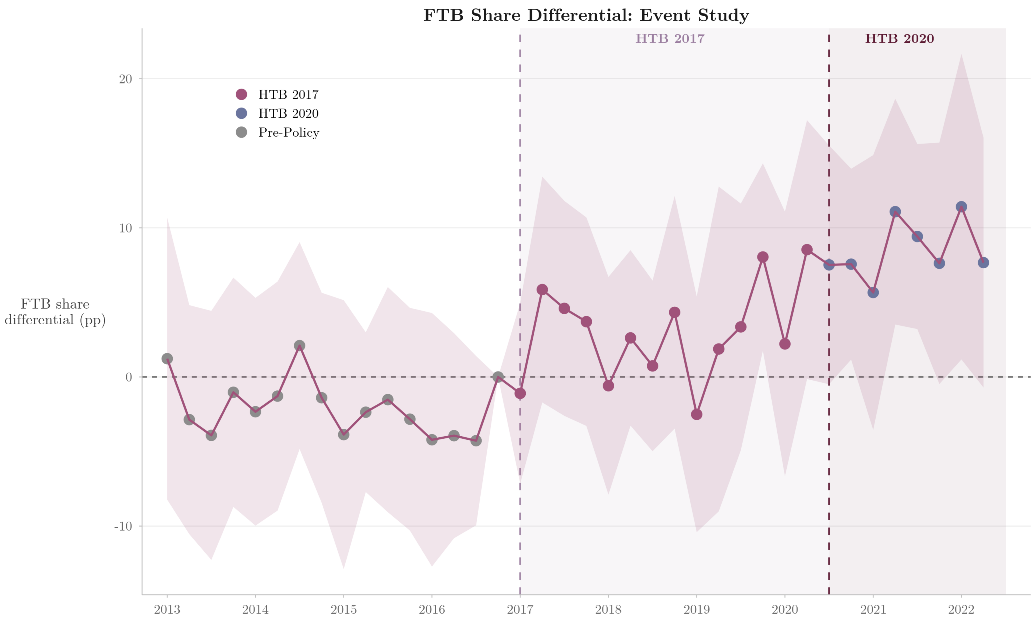

Figure 13: Quarterly event study: differential first-time buyer share (new-build minus second-hand), with 95% confidence intervals. Reference period is 2016Q4. Standard errors clustered by RPPI region.

Figure 13 reports the corresponding quarterly event study. The pre-treatment coefficients are small and individually insignificant throughout 2013–2016. The joint Wald test likewise does not reject the null of no pre-treatment differential movement, $F(15, 32) = 1.014$, $p = 0.4661$. This non-rejection should not be treated as definitive proof, but in conjunction with the raw series it is consistent with the identifying assumption used in the main difference-in-differences specification. After 2017, the event-study coefficients are mostly positive and increase further around the enhanced regime, although they are not estimated with high precision.

8.4.2 New-Build Share: First-Time Buyers versus Former Owner-Occupiers

I next turn to the design that compares the new-build share for first-time buyers and former owner-occupiers. Here, the identifying assumption is that, absent HTB, the two buyer groups would have followed parallel trends in their relative concentration in the new-build segment. Figure 14 again provides a first diagnostic using the raw annual series.

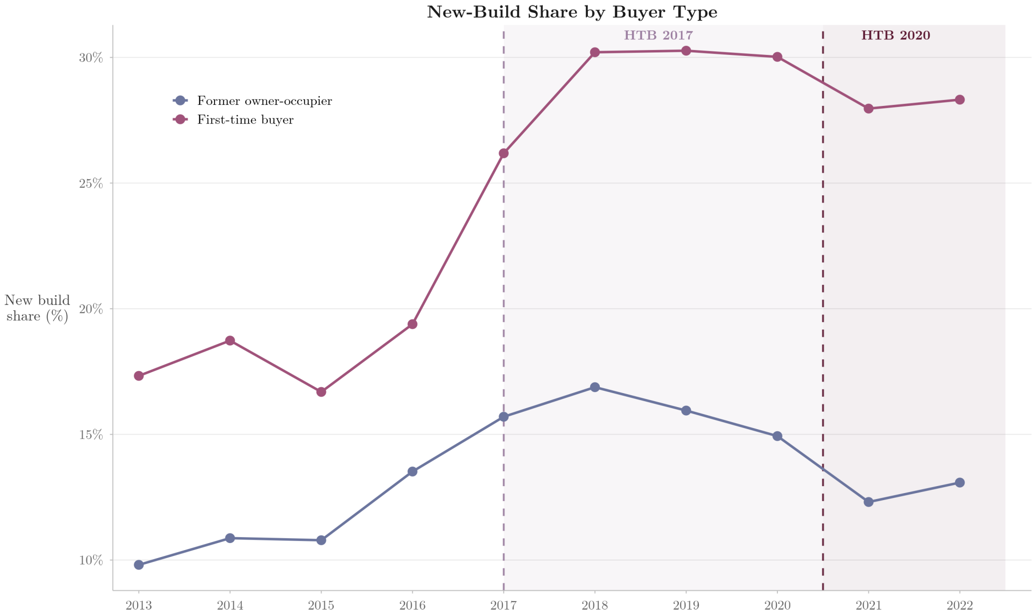

Figure 14: New-build share of transactions by buyer type, annual.

Figure 14 plots the new-build share separately for first-time buyers and former owner-occupiers. Both series rise during the pre-policy market recovery, but the gap between them does not display a clear pre-policy widening. After HTB was introduced in 2017, the first-time buyer series increases much more strongly than the former owner-occupier series.

To examine pre-treatment dynamics more formally, I estimate the corresponding quarterly event-study specification:

$$Share^{NB}_{rbt} = \mu_{rb} + \sum_{q \neq 2016Q4} \alpha_q \mathbf{1}\{t \in q\} + \sum_{q \neq 2016Q4} \delta_q \big(FTB_b \times \mathbf{1}\{t \in q\}\big) + v_{rbt}, \quad (21)$$

where μ_{rb} denotes a fixed effect for each region $\times$ buyer-type cell and standard errors are again clustered by RPPI region. The coefficients δ_q trace the evolution of the differential new-build share between first-time buyers and former owner-occupiers. The corresponding Wald test evaluates the joint null that all pre-2017 interaction coefficients are equal to zero.

Figure 15 reports the quarterly event study. The pre-treatment coefficients are again mostly small and individually insignificant. The joint Wald test does not reject the null of no pre-treatment differential movement, $F(15, 32) = 1.084$, $p = 0.4074$. As with the first design, this does not prove the identifying assumption, but it is consistent with the difference-in-differences interpretation once read together with the raw series. After 2017, the event-study coefficients are mostly positive and increase further around

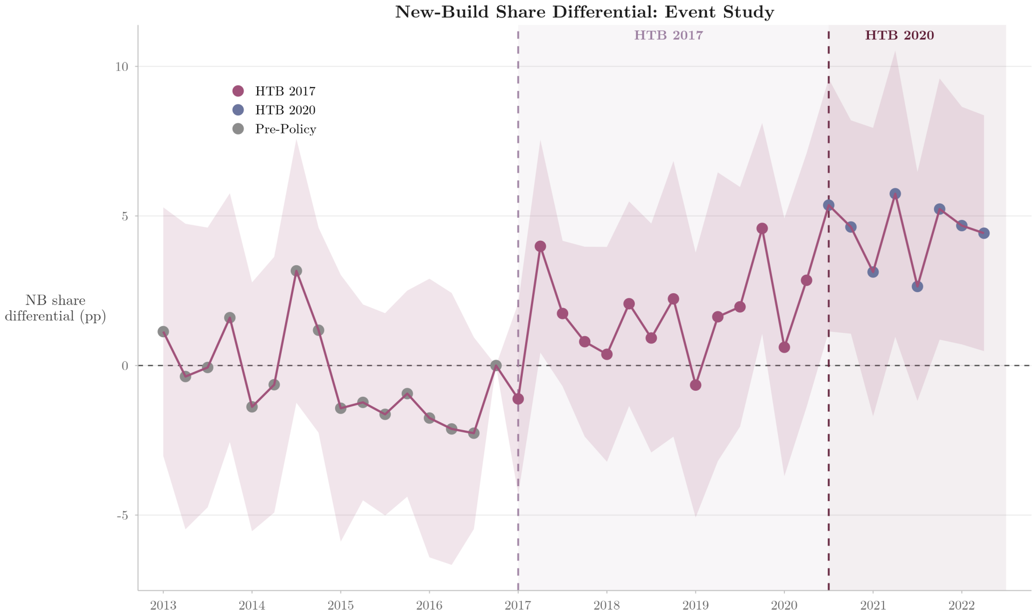

Figure 15: Quarterly event study: differential new-build share (first-time buyers minus former owner-occupiers), with 95% confidence intervals. Reference period is 2016Q4. Standard errors clustered by RPPI region.

the enhanced regime, although they are not estimated with high precision.

The raw plots, event-study profiles, and Wald tests provide no indication that the relevant treatment-comparison differentials were already widening before HTB was introduced. The post-2017 divergence in both designs therefore appears in the data only once the policy creates a reason for it to emerge, which supports a causal interpretation of both difference-in-differences estimates.

9 Conclusion

This paper has studied how a targeted housing purchase subsidy reshapes behaviour on two margins simultaneously: locally, at the eligibility boundary, and compositionally, within the treated segment. The theoretical framework links those margins through a single object, namely the share of demand exposed to the notch, and the empirical analysis shows that both respond in tandem to subsidy generosity. Using Ireland's Help to Buy scheme as its setting, and combining a developer-side notch model with bunching estimation, a distributional difference-in-differences, and two buyer-participation difference-in-differences designs, the analysis has identified two margins of response connected by a single underlying mechanism.

The first margin is the local distortion at the €500,000 eligibility ceiling. Bunching estimation reveals excess mass just below the threshold and missing mass above it in the new-build market during both policy regimes, with no comparable pattern in the second-hand market or in the pre-policy period. Under the polynomial counterfactual, the normalised bunching coefficient is $\hat{b} = 1.352$ under the initial regime and 2.090 under the enhanced regime, implying a transaction-value response of €6,759 and €10,448 for the marginal developer who relocates below the threshold, around one-third of the maximum statutory relief in each regime. Both quantities move with subsidy generosity between regimes: the bunching coefficient is 55% higher against a 50% rise in maximum statutory relief, and the implied transaction-value response grows alongside τ. The distributional difference-in-differences confirms that the response is sharply local to the cutoff rather than diffuse across the wider transaction-value distribution, and is stronger under the enhanced regime.

Because $v = p \cdot h$, these magnitudes reflect adjustment in the transacted bundle rather than in the unit price alone. The alternative counterfactual in Appendix B preserves the monotonic increase in $\hat{b}$ with subsidy generosity, though the precise cross-regime proportionality is specific to the polynomial specification. Read in terms of the framework of Section 4, the cross-regime strengthening of the threshold response is consistent with both channels of $dE(\tau, \lambda)/d\tau$, direct valuation and participation, moving in the same direction as τ rises.

The second margin is the composition of demand in the treated segment. Relative to second-hand purchases, the first-time buyer share of new-build purchases rises by 5.11 percentage points under the initial regime and 10.50 percentage points under the enhanced regime. Relative to former owner-occupiers, the new-build share of first-time buyer purchases rises by 2.48 and 5.33 percentage points respectively. Across both designs, the buyer-participation response scales closely with the doubling of the marginal subsidy rate from 5% to 10%, consistent with an approximately linear local response over the range of subsidy generosity observed in this setting. The buyer-participation response therefore moves with subsidy generosity in the same direction as the threshold response, which is what the theoretical framework predicts when the value of eligibility and the eligible share of demand both increase simultaneously. In this sense, a targeted subsidy does not merely reallocate transactions around its eligibility boundary but also alters who participates in the treated market, and the theoretical framework I develop makes that link explicit.

These findings make three contributions: the first market-level analysis of Ireland's Help to Buy scheme, complementing the borrower-level evidence in McCann and Singh (2023); an extension of the housing-notch literature (for example, Best and Kleven, 2018; Kopczuk and Munroe, 2015; Agarwal, Hu, and Lee, 2023) to the partial-coverage case in which only a subset of demand faces the discontinuity; and evidence on how an explicitly targeted subsidy reshapes buyer composition across market segments, adding to a literature that has typically studied unintentional displacement (Carozzi, 2020; Berger, Turner, and Zwick, 2020; Lambie-Hanson, Li, and Slonkosky, 2022).

Several limitations temper these conclusions. The first set concerns what the data can measure. The Property Price Register contains no hedonic property characteristics, so the implied transaction-value response cannot be decomposed into adjustments in the unit price (p) and housing quality (h). Settings with richer data, such as those in Agarwal, Hu, and Lee (2023) and Uribe (2022), have shown that notches in tax and subsidy schedules can induce developers to deliver smaller units, and whether a similar channel operates in this setting remains an open question. A related concern, raised by C. K. Y. Leung, T. C. Leung, and Tsang (2015) in the context of stamp-duty notches in Hong Kong, is that bunching at a price ceiling may in principle reflect unrecorded side payments that leave the recorded value below the threshold whilst the true exchange value remains above it. This channel cannot be ruled out without hedonic data, but it is a general feature of housing-notch identification.

The second set concerns the reach of the sample. The estimation window ends in July 2022 to avoid contamination from the First Home Scheme, so any longer-run dynamics, including the possibility that the buyer-participation response attenuates or intensifies under prolonged policy exposure, lie beyond what this analysis can assess. The First Home Scheme itself offers a particularly promising setting for further study: its transaction-value ceilings vary by county rather than applying at a single national threshold, which would allow within-country variation in the location of the notch to be exploited directly, though credibly comparing properties on either side of those county-specific ceilings would again require hedonic data. Finally, a structural model linking the threshold and buyer-participation margins could quantify the welfare effects of the eligibility ceiling and evaluate counterfactual designs, including the replacement of the sharp eligibility cutoff with a tapered reduction in the subsidy rate.

For policy, the results point to a specific trade-off in the design of subsidies with eligibility cutoffs. Narrowing the eligible segment, whether by buyer group, dwelling type, or price ceiling, does not eliminate the distortion that a targeted subsidy creates at its boundary. Instead, it concentrates that distortion at a specific price point and in a specific part of the market. At the same time, targeting succeeds in drawing additional members of the intended buyer group into the treated segment, and this compositional response also strengthens with subsidy generosity. Whether this rise in first-time buyer-participation reflects a real expansion of access to homeownership or instead the acceleration of purchases that would have occurred anyway is a question that my data cannot fully resolve. For the English HTB scheme, Boileau, Conwell, and Levell (2026) find that simulated affordability gains were concentrated among higher-income households who could likely have bought without the subsidy, though a shared-equity loan and a tax rebate are sufficiently different instruments that the Irish answer need not be the same. Settling it would require matched buyer-level microdata, and the answer matters: the case for HTB rests in part on whether it creates new homeowners or merely advances the timing of purchases that would have happened anyway.

A second open question is whether the eligibility threshold generates allocative distortions of the kind documented by Hargaden (2017), whose Irish pre-2007 stamp-duty exemption generated bunching of eligible buyers below the threshold and bunching of ineligible buyers above it, with the latter outbidding the former in the above-threshold range. In this setting, where developers set prices rather than bid against one another, the analogous concern operates across both segments of the market. Within the new-build segment, HTB may funnel first-time buyers into a particular subset of units positioned just below the cutoff, which may differ in quality, size, or location from the units those buyers would otherwise have chosen. Across segments, the buyer-participation evidence shows that HTB also draws first-time buyers out of the second-hand market and into new-builds. The welfare effects of the scheme should therefore be evaluated not only in terms of homeownership access but also in terms of the dwelling-type margin along which that access shifts.

Appendix A Additional Tables

Table 11: Density Discontinuity

	Pre-HTB		HTB 2017		HTB 2020	
	Raw	Jittered	Raw	Jittered	Raw	Jittered
Panel A: New Build						
$\hat{f}^-$	0.96	1.21	2.29	1.98	2.78	2.17
$\hat{f}^+$	0.77	0.90	0.45	0.88	0.12	1.59
T	-0.901	-1.141	-11.274***	-7.209***	-11.458***	-2.097**
p-value	0.368	0.254	0.000	0.000	0.000	0.036
N_{left}	8,915	8,907	24,550	24,462	14,213	14,130
N_{right}	633	641	1,895	1,983	1,153	1,236
Panel B: Second Hand						
$\hat{f}^-$	0.22	0.81	0.48	0.82	0.66	1.19
$\hat{f}^+$	0.57	0.96	0.55	0.78	0.57	1.03
T	3.363***	1.405	0.673	-0.596	-0.738	-1.284
p-value	0.001	0.160	0.501	0.551	0.460	0.199
N_{left}	46,921	47,083	66,851	67,070	47,641	47,818
N_{right}	3,498	3,336	5,121	4,902	4,247	4,070

Notes: Cattaneo, Jansson, and Ma (2018). MSE-optimal bandwidth; $p = 2$. Jittered: uniform noise ±€2,500. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 12: Bunching Estimates: Polynomial Order Sensitivity

		$p = 4$	$p = 5$	$p = 6$	$p = 7$
Panel A: New Build					
HTB 2017	$\hat{b}$	3.048**	**1.352***	1.739*	1.745*
		[0.201, 12.066]	[-0.229, 4.685]	[-0.046, 7.879]	[-0.051, 7.889]
HTB 2020	$\hat{b}$	2.697*	**2.090***	2.183*	2.090*
		[-0.092, 11.831]	[-0.231, 9.343]	[-0.383, 13.578]	[-0.392, 13.225]
Panel B: Second Hand					
HTB 2017	$\hat{b}$	0.065	**0.091**	0.037	0.045
		[-0.635, 0.836]	[-0.624, 0.886]	[-0.647, 0.810]	[-0.662, 0.823]
HTB 2020	$\hat{b}$	0.210	**0.307**	0.192	0.184
		[-0.438, 0.996]	[-0.397, 1.176]	[-0.505, 1.032]	[-0.509, 1.037]

Notes: Baseline specification in bold ($p = 5$). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 13: Bunching Estimates: Bin-Width Sensitivity

		€5k	€10k
Panel A: New Build			
HTB 2017	$\hat{b}$	**1.352***	0.053
		[-0.229, 4.685]	[-0.658, 1.113]
HTB 2020	$\hat{b}$	**2.090***	0.594
		[-0.231, 9.343]	[-0.641, 3.040]
Panel B: Second Hand			
HTB 2017	$\hat{b}$	**0.091**	-0.251
		[-0.624, 0.886]	[-0.652, 0.141]
HTB 2020	$\hat{b}$	**0.307**	-0.094
		[-0.397, 1.176]	[-0.485, 0.323]

Notes: Baseline specification in bold (€5,000 bins). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 14: Bunching Estimates: Exclusion Zone Sensitivity

		1 bin	2 bins	3 bins
Panel A: New Build				
Pre-Policy	$\hat{b}$	**-0.417**	-0.447	-0.804
		[-2.020, 1.053]	[-2.342, 1.899]	[-2.943, 2.223]
HTB 2017	$\hat{b}$	**1.352***	0.969	0.873
		[-0.229, 4.685]	[-0.807, 4.423]	[-1.150, 4.835]
HTB 2020	$\hat{b}$	**2.090***	1.568	2.335
		[-0.231, 9.343]	[-0.936, 8.753]	[-1.057, 15.685]
Panel B: Second Hand				
Pre-Policy	$\hat{b}$	**-0.177**	-0.307	-0.123
		[-0.873, 0.626]	[-1.211, 0.899]	[-1.397, 1.544]
HTB 2017	$\hat{b}$	**0.091**	-0.078	0.614
		[-0.624, 0.886]	[-0.902, 0.978]	[-0.687, 2.493]
HTB 2020	$\hat{b}$	**0.307**	0.182	0.973
		[-0.397, 1.176]	[-0.703, 1.383]	[-0.308, 2.810]

Notes: Baseline specification in bold (1 bin excluded below notch). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 15: Bunching Estimates: Bootstrap Replication Sensitivity

		$B = 500$	$B = 1{,}000$	$B = 1{,}500$
Panel A: New Build				
Pre-Policy	$\hat{b}$	-0.417	**-0.417**	-0.417
		[-1.993, 0.927]	[-2.020, 1.053]	[-1.915, 1.109]
HTB 2017	$\hat{b}$	1.352*	**1.352***	1.352*
		[-0.150, 4.600]	[-0.229, 4.685]	[-0.185, 4.488]
HTB 2020	$\hat{b}$	2.090*	**2.090***	2.090*
		[-0.112, 7.518]	[-0.231, 9.343]	[-0.223, 8.684]
Panel B: Second Hand				
Pre-Policy	$\hat{b}$	-0.177	**-0.177**	-0.177
		[-0.862, 0.588]	[-0.873, 0.626]	[-0.891, 0.626]
HTB 2017	$\hat{b}$	0.091	**0.091**	0.091
		[-0.633, 0.880]	[-0.624, 0.886]	[-0.660, 0.954]
HTB 2020	$\hat{b}$	0.307	**0.307**	0.307
		[-0.373, 1.183]	[-0.397, 1.176]	[-0.424, 1.151]

Notes: Baseline specification in bold ($B = 1{,}000$ bootstrap replications). 95% percentile bootstrap CIs in brackets. Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Appendix B Alternative Counterfactual

This appendix reports an alternative bunching exercise that uses second-hand transactions to construct the counterfactual new-build distribution[22]. The baseline polynomial estimator recovers the untreated distribution from a smooth function fitted to the new-build data itself. That approach is standard, but it necessarily ties the counterfactual to a functional-form approximation of the treated distribution. Here, by contrast, I use an external untreated benchmark. Because second-hand properties were never eligible for HTB, their transaction-value distribution provides a natural reference point for how new-build transactions might have evolved absent the policy.

To construct the second-hand counterfactual, I model the no-policy new-build distribution as a function of the second-hand distribution and the smooth shape of the transaction-value density. Let $c_i^{NB,0}$ and $c_i^{SH,0}$ denote, respectively, the numbers of new-build and second-hand transactions in bin i during the pre-policy baseline period, where transaction-values are grouped into €5,000 bins.[23] Let z_i denote the distance between the midpoint of bin i and the eligibility threshold at $\bar{v} =$ €500,000. I estimate

$$c_i^{NB,0} = \alpha + \delta\, c_i^{SH,0} + \sum_{j=1}^{q} \beta_j (z_i)^j + \sum_{r\in R} \eta_r \mathbf{1}\left\{\frac{\bar{v} + z_i}{r} \in \mathbb{N}\right\} + e_i, \tag{22}$$

where $q = 5$, $R = \{10{,}000, 25{,}000, 50{,}000\}$, $\mathbb{N}$ denotes the set of natural numbers, and $\mathbf{1}\{\cdot\}$ is the indicator function. The second-hand term captures the extent to which variation in second-hand counts predicts variation in new-build counts across bins, whilst the polynomial and round-number terms flexibly absorb the smooth shape of the transaction-value distribution and focal-value heaping.

I then construct the treatment-period counterfactual by replacing baseline second-hand counts with observed second-hand counts from the relevant treatment period:

$$\hat{c}_i^{0,NB}(t) = \hat{\alpha} + \hat{\delta}\, c_i^{SH}(t) + \sum_{j=1}^{q} \hat{\beta}_j (z_i)^j + \sum_{r\in R} \hat{\eta}_r \mathbf{1}\left\{\frac{\bar{v} + z_i}{r} \in \mathbb{N}\right\}, \tag{23}$$

and I rescale $\hat{c}_i^{0,NB}(t)$ so that total predicted new-build transactions match the observed total in treatment period t. Excess bunching $\hat{B}$, missing mass $\hat{M}$, and the normalised bunching coefficient $\hat{b}$ are then computed exactly as in equations (12)–(14),

[22] The logic of this estimator is not dissimilar to Cengiz et al. (2019), who use an untreated group's distribution as the counterfactual in a distributional difference-in-differences framework, rather than relying on a within-group polynomial approximation.

[23] As in the polynomial bunching estimator, transactions priced at exactly €500,000 are assigned to the [€500,000, €505,000) bin rather than to the final eligible bin [€495,000, €500,000), for the reasons discussed in Section 6.1.

with the second-hand-based counterfactual replacing the polynomial counterfactual.

The identifying assumption is that, absent HTB, the conditional relationship between new-build and second-hand bin counts would have remained stable over time, after accounting for the polynomial and round-number controls. Inference is based on a bootstrap that resamples underlying transaction-values in both the baseline and treatment periods. As in the polynomial specification, I report percentile bootstrap confidence intervals and bootstrap p-values.

B.1 Treatment Estimates

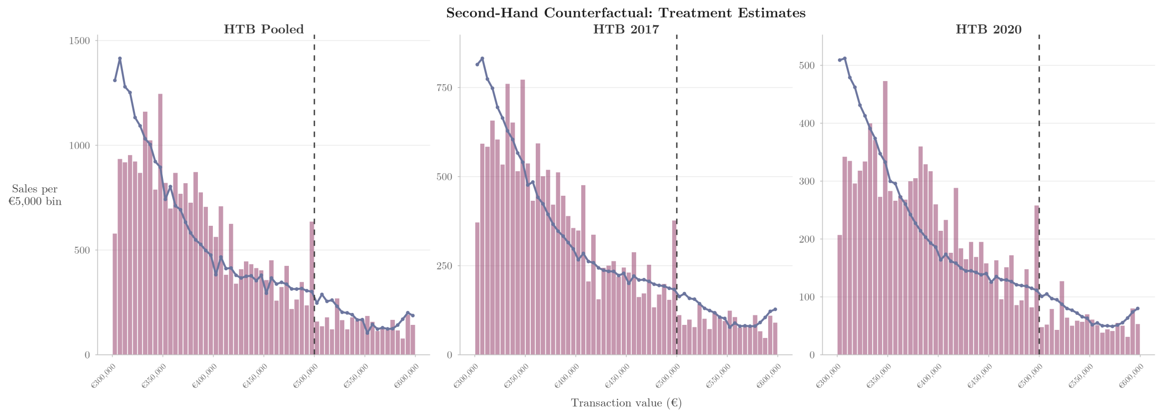

Figure 16: Observed new-build bin counts (bars) and second-hand-based counterfactual (line), by regime. Dashed line marks the €500,000 notch.

Figure 16 plots the observed new-build distribution against the second-hand-based counterfactual for the three treatment samples. In each panel, the observed distribution displays clear excess mass in the final eligible bin, [€495,000, €500,000), and missing mass above the threshold extending roughly four bins into the ineligible region, from [€500,000, €505,000) to [€515,000, €520,000). This local pattern is similar to that obtained under the baseline polynomial counterfactual. Although the fit is less exact away from the notch, the second-hand-based counterfactual passes smoothly through the threshold region and tracks the distribution closely over [€450,000, €550,000), where the bunching response is concentrated.

Table 16 reports the corresponding estimates. The normalised bunching coefficient is $\hat{b} = 1.055$ under the initial regime and $\hat{b} = 1.316$ under the enhanced regime, both significant at the 1% level. These estimates are smaller than the polynomial-counterfactual estimates in Table 4 (1.352 and 2.090), particularly under the enhanced regime, but they again confirm prediction (i) of the model: substantial excess mass below the threshold, with a stronger normalised response under the more generous

Table 16: Second-Hand Counterfactual: Treatment Estimates

	HTB 2017	HTB 2020
$\hat{b}$	1.055***	1.316***
	[0.798, 1.343]	[0.995, 1.674]
$\hat{B}$	193.6	146.6
	[152.6, 236.0]	[113.9, 177.9]
$\hat{M}$	52.5	52.8
$\hat{B}/\hat{M}$	3.69	2.78
Baseline	2013–16	2013–16
Treatment	2017–20	2020–22
Polynomial	5	5

Notes: Counterfactual from Eq. (22). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

regime, in line with prediction (ii) and the comparative static on τ in Equation (2). A further notable feature is that the second-hand-based confidence intervals exclude zero under both regimes, whereas the polynomial intervals straddle it, so that the second-hand counterfactual delivers materially tighter inference at the cost of somewhat smaller point estimates. The lower level of $\hat{B}$ under the enhanced regime again reflects the shorter sample window rather than a weaker behavioural response.

The cross-regime scaling of the bunching coefficient is, however, somewhat weaker under the second-hand counterfactual than under the polynomial specification. Under the polynomial counterfactual, $\hat{b}$ rises by approximately 55% between regimes against a 50% increase in the maximum statutory relief, which Section 7.1 reads as evidence that the local response of bunching intensity strengthens with τ. Under the second-hand counterfactual, the corresponding increase is from 1.055 to 1.316, or about 25%, which is meaningfully smaller than the 50% rise in maximum statutory relief.

This attenuation follows from how the two estimators handle common market-level variation. The polynomial counterfactual is fitted only from new-build data, so any shock to the new-build distribution near the notch is projected through the smooth polynomial and attributed to the threshold response. The second-hand counterfactual, by contrast, absorbs variation that is common to both markets into the coefficient δ on $c_i^{SH}(t)$. If any of the cross-regime increase in $\hat{b}$ under the polynomial specification reflects a shock that moves the new-build and second-hand distributions together, rather than the HTB threshold specifically, the second-hand estimator will partial that component out and return a smaller cross-regime ratio. A related possibility is that the polynomial counterfactual over-fits the cross-regime difference by projecting a smooth function across the notch in a way that the external benchmark does not. Either reading leaves the direction of the comparative static on τ intact, but indicates that the magnitude of the proportional scaling is less precisely identified than the baseline specification alone would suggest.

The second-hand counterfactual also places the Irish estimates more uniformly within the partial-coverage region of the housing-notch literature. Under the polynomial counterfactual, the initial-regime $\hat{b} = 1.352$ lies below the 1.64–1.85 range reported by Best and Kleven (2018), whereas the enhanced-regime $\hat{b} = 2.090$ exceeds it. Under the second-hand counterfactual, both regime estimates lie below the Best and Kleven (2018) range, and both remain substantially smaller than the 8.39 reported by Agarwal, Hu, and Lee (2023) for a universal-notch setting. This pattern gives a cleaner illustration of the $\lambda < 1$ attenuation logic developed in Section 4: when demand exposed to the notch is only a fraction of total new-build demand, the bunching response at the cutoff is mechanically dampened relative to universal-notch benchmarks, and this holds across both HTB regimes rather than only under the initial schedule.

The main difference relative to the polynomial counterfactual lies in the estimated missing mass above the threshold and, relatedly, in the implied $\hat{B}/\hat{M}$ ratio. Under the polynomial counterfactual in Table 4, missing mass exceeds excess mass in both regimes ($\hat{B}/\hat{M} = 0.42$ and 0.81), whereas under the second-hand counterfactual the reverse is true: $\hat{B}/\hat{M} = 3.69$ under the initial regime and 2.78 under the enhanced regime. The excess-mass estimates below the threshold are broadly similar across the two approaches. The divergence comes primarily from how the counterfactual is fitted in the region above the notch. This suggests that the precise decomposition of the response into excess mass below and missing mass above is sensitive to the choice of counterfactual, even though the existence of substantial bunching below €500,000 is not.

The direction of the $\hat{B}/\hat{M}$ flip is again informative. Under the second-hand counterfactual, the Irish pattern moves closer to that reported by Agarwal, Hu, and Lee (2023) for New South Wales, where excess mass exceeds missing mass and is interpreted as evidence of net entry into the subsidised segment. The main-text polynomial results are not consistent with a pure net-entry interpretation, but the buyer-participation evidence in Section 7.4 shows that HTB does draw additional first-time buyers into the new-build segment, raising the effective share λ of demand exposed to the notch. The second-hand counterfactual therefore produces a mass decomposition that aligns more naturally with the participation channel identified by the buyer-participation designs than the polynomial specification does, even though the first-stage existence of threshold bunching is robust across both constructions.

The second-hand counterfactual therefore strengthens the core empirical conclusion of the paper without materially changing its economic interpretation. It confirms that the treated new-build distribution displays substantial excess mass below the HTB threshold and that the normalised response is larger under the enhanced regime, in line with prediction (ii). At the same time, the appendix indicates that the exact balance between excess mass below and missing mass above the threshold, and the precise proportionality of $\hat{b}$ to τ across regimes, are both less robust than the existence of bunching itself. For that reason, I place greater weight on the bunching coefficient $\hat{b}$ and the implied transaction-value response, which are more stable in sign and significance across counterfactual constructions, than on the cross-regime scaling factor or the $\hat{B}/\hat{M}$ ratio, which depend on how the counterfactual is fitted in the region above the notch.

B.2 Placebo Estimates

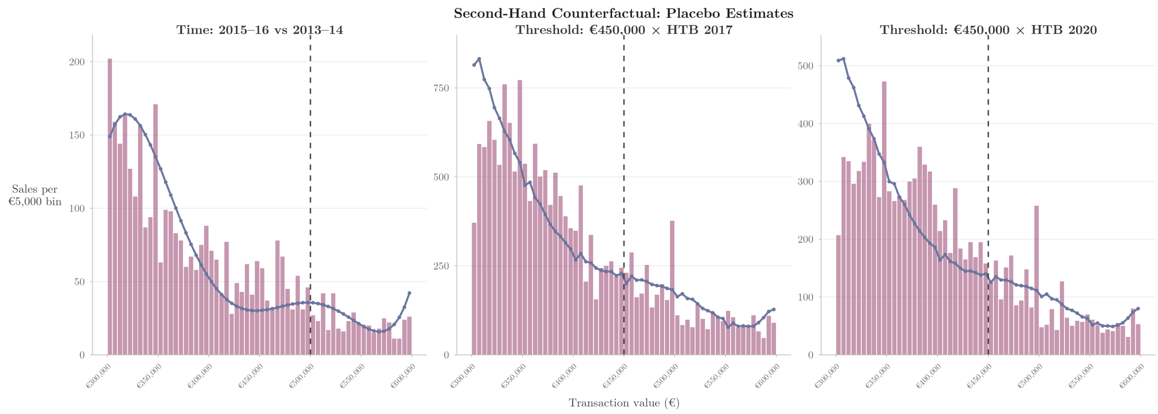

Figure 17: Second-hand-based counterfactual: placebo estimates. Left panel: time placebo (2015–16 vs 2013–14 baseline at €500,000). Centre and right panels: falsification exercise with threshold at €450,000. Dashed lines mark the relevant threshold.

Figure 17 plots the observed new-build distribution against the second-hand-based counterfactual in the three placebo exercises. As in the treatment figures, the fit is less exact away from the threshold region: the counterfactual tends to lie above the observed distribution at the lower end of the window and below it in the middle of the distribution. The time placebo exhibits a smoother counterfactual line than the treatment panels because the 2013–2014 baseline contains fewer new-build transactions, giving the polynomial terms greater weight relative to the second-hand regressor. Around the relevant threshold, however, the fit is close, and none of the placebo panels displays the pronounced excess mass visible at the true €500,000 notch in the treatment estimates.

Table 17: Second-Hand Counterfactual: Placebo Estimates

	Time	€450,000 × HTB 2017	€450,000 × HTB 2020
$\hat{b}$	0.290 [-0.096, 0.748]	0.072 [-0.081, 0.254]	0.126 [-0.054, 0.323]
$\hat{B}$	10.3	16.4	17.7
$\hat{M}$	8.6	-30.4	-0.4
$\hat{B}/\hat{M}$	1.21	-0.54	-44.25
Threshold	€500,000	€450,000	€450,000
Baseline	2013–14	2013–16	2013–16
Treatment	2015–16	2017–20	2020–22

Notes: 95% bootstrap percentile CIs (1,000 replications). $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$. Negative M and B/M indicate no missing mass above the fake threshold.

The placebo exercises in this appendix are chosen to match the identifying assumption of the second-hand counterfactual. In the main polynomial bunching analysis, the natural placebo checks are time and type placebos, because the counterfactual is recovered from the treated distribution by functional-form extrapolation. Here, by contrast, the counterfactual is constructed from the relationship between new-build and second-hand transactions. The relevant question is therefore whether that relationship is stable over time and whether the method spuriously detects bunching at a salient threshold where no policy discontinuity exists.

I therefore report one time placebo and two threshold placebos. The time placebo uses 2013–2014 as the baseline and 2015–2016 as a pseudo-treatment period at the true €500,000 threshold. The threshold placebos re-centre the exercise at €450,000 during the initial and enhanced HTB periods. I choose €450,000 because it is a salient round-number in the same part of the transaction-value distribution as the true notch, but remains sufficiently far from the €400,000 kink in the initial regime and the €300,000 kink in the enhanced regime to avoid contamination from those slope changes.

Table 17 reports the results. In the time placebo, the estimated bunching coefficient at the true €500,000 threshold is 0.290 and statistically insignificant. This supports the assumption that the pre-policy relationship between new-build and second-hand bin counts is stable over time. The two threshold placebos are likewise null. At the falsified €450,000 threshold, the estimated bunching coefficients are 0.072 in the initial regime and 0.126 in the enhanced regime, and neither estimate is statistically significant. The method therefore does not spuriously detect excess mass at a salient round-number threshold where no HTB discontinuity exists.

These placebo results reinforce the treatment estimates. The second-hand counterfactual yields null findings in a pre-policy pseudo-treatment period and at a false threshold, but detects clear excess mass at the true €500,000 notch during the HTB period. This is consistent with the null placebo evidence from the main polynomial bunching analysis, despite the different counterfactual constructions, and suggests that the estimated treatment pattern is not an artefact of the method.

B.3 Implied Transaction-Value Response

Table 18 reports the implied transaction-value responses under the second-hand counterfactual. The implied response is $\widehat{\Delta v^*}$ = €5,275 under the initial regime and $\widehat{\Delta v^*}$ = €6,580 under the enhanced regime, corresponding to 26.4% and 21.9% of the maximum statutory relief respectively.

Table 18: Second-Hand Counterfactual: Implied Transaction-Value Response

	HTB 2017	HTB 2020
$\hat{b}$	1.055***	1.316***
	[0.798, 1.343]	[0.995, 1.674]
$\widehat{\Delta \bar{v}}$	€5,275	€6,580
	[€3,990, €6,715]	[€4,975, €8,370]
$\widehat{\Delta \bar{v}}/\tau$	26.4%	21.9%
	[20.0%, 33.6%]	[16.6%, 27.9%]
τ	€20,000	€30,000
Bin width	€5,000	€5,000

Notes: Second-hand counterfactual from Eq. (22). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Relative to the polynomial-counterfactual estimates in Table 5, the second-hand counterfactual responses are smaller, in line with the smaller bunching coefficients reported above. The direction of the comparative static on τ in prediction (iii) is preserved: the implied transaction-value response remains positive under both regimes and is larger under the more generous schedule. As in the baseline specification, the implied adjustment remains well below the full subsidy, which is consistent with the marginal buncher's indifference condition in Equation (7). Since $v = p \cdot h$, the estimate should again be interpreted as adjustment in the transacted bundle rather than as a pure reduction in the unit price.

One notable difference relative to the baseline specification concerns the ratio $\widehat{\Delta v^*}/\tau$. Under the polynomial counterfactual, this ratio is around one-third in both regimes, which Section 7.2 reads as evidence that the marginal buncher's response strengthens with the value of eligibility. Under the second-hand counterfactual, the ratio falls from 26.4% to 21.9% as τ rises, so the response still grows in absolute terms but does not scale proportionally. As with the cross-regime pattern in the bunching coefficient itself, the near-exact proportionality that the baseline specification delivers is therefore partly a feature of the polynomial counterfactual rather than a robust property of the data. The direction of prediction (iii) is preserved, but the proportionality of the response to τ is not itself robust to the choice of counterfactual.

The second-hand counterfactual also yields substantially tighter confidence intervals than the polynomial approach. For the initial regime, the 95% confidence interval is [€3,990,€6,715], compared with [€-1,147,€23,427] under the polynomial counterfactual. For the enhanced regime, the corresponding intervals are [€4,975,€8,370] and [€-1,156,€46,713]. In both regimes, the second-hand-based intervals exclude zero, whereas the polynomial intervals straddle it. Using an external untreated benchmark therefore affects precision as well as point estimates, and strengthens the conclusion that HTB induced a positive and economically meaningful transaction-value response.

B.3.1 Bin-Width Sensitivity

Table 19: Second-Hand Counterfactual: Implied Transaction-Value Response (Bin-Width Sensitivity)

Regime	Bin width	$\hat{b}$	$\widehat{\Delta \bar{v}}$	$\widehat{\Delta \bar{v}}/\tau$
HTB 2017	€2,500	1.268***	€3,170	15.8%
			[€2,410, €3,972]	[12.0%, 19.9%]
	€5,000	1.055***	€5,275	26.4%
			[€3,990, €6,715]	[20.0%, 33.6%]
	€10,000	0.420***	€4,200	21.0%
			[€2,110, €6,380]	[10.5%, 31.9%]
HTB 2020	€2,500	2.225***	€5,562	18.5%
			[€4,368, €6,840]	[14.6%, 22.8%]
	€5,000	1.316***	€6,580	21.9%
			[€4,975, €8,370]	[16.6%, 27.9%]
	€10,000	0.521***	€5,210	17.4%
			[€3,100, €7,400]	[10.3%, 24.7%]

Notes: Second-hand counterfactual from Eq. (22). 95% percentile bootstrap CIs in brackets (1,000 replications). Bootstrap p-values: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 19 examines the sensitivity of the second-hand counterfactual transaction-value response to the choice of bin width[24]. The implied response remains positive and significant at the 1% level across all three bin widths in both regimes, and the enhanced-regime estimate exceeds the initial-regime estimate throughout. This contrasts with the polynomial counterfactual, where the €10,000 bin-width estimates in Table 9 fall to €530 and €5,938 and lose statistical significance entirely. As under the polynomial approach, the exact level varies somewhat with bin width because the fitted counterfactual changes with bin resolution, but the variation is more contained and does not threaten inference. The improvement in bin-width robustness under the second-hand counterfactual reflects the greater structure imposed by the external untreated benchmark in the region above the threshold, where the polynomial specification is most dependent on functional-form extrapolation and where, in the theoretical framework of Section 4, the dominated region above $\bar{v}$ is located. The second-hand counterfactual therefore reinforces the main result that the implied transaction-value response is positive, larger under the enhanced scheme, and not materially sensitive to the precise binning choice.

[24] Because the second-hand counterfactual does not require a two-pass notch-width procedure, the right-side bin constraint that limits the polynomial estimator to €5,000 and €10,000 bins does not bind here. The €2,500 specification is therefore feasible: the counterfactual is constructed from a regression on second-hand counts rather than from the shape of the threshold bin itself, so the narrow bin width does not produce the degenerate estimates observed under the polynomial approach.

References

Agarwal, Sumit, Maggie Hu, and Adrian D Lee (2023). "Market Impacts and Unintended Consequences of Housing Assistance Policies with Price Threshold". In: *Working Paper*. URL: https://doi.org/10.2139/ssrn.4612792.

Anagol, Santosh et al. (2024). "Diffuse Bunching with Frictions: Theory and Estimation". In: *SSRN Electronic Journal*. URL: https://doi.org/10.2139/ssrn.4874241.

Berger, David, Nicholas Turner, and Eric Zwick (2020). "Stimulating Housing Markets". In: *The Journal of Finance* 75.1, pp. 277–321. URL: https://doi.org/10.1111/jofi.12847.

Best, Michael Carlos and Henrik J. Kleven (2018). "Housing Market Responses to Transaction Taxes: Evidence from Notches and Stimulus in the U.K." In: *The Review of Economic Studies* 85.1, pp. 157–193. URL: https://doi.org/10.1093/restud/rdx032.

Boileau, Bee, Lucas Conwell, and Peter Levell (2026). *Who is Helped by Help to Buy Schemes?* IFS Working Paper 26/27. Institute for Fiscal Studies. URL: https://ifs.org.uk/sites/default/files/2026-04/WP202627-Who-is-helped-by-help-to-buy-schemes_0.pdf.

Carozzi, Felipe (2020). "Credit Constraints and the Composition of Housing Sales. Farewell to First-Time Buyers?" In: *Journal of the European Economic Association* 18.3. URL: https://doi.org/10.1093/jeea/jvz017.

Carozzi, Felipe, Christian A. L. Hilber, and Xiaolun Yu (2024). "On the Economic Impacts of Mortgage Credit Expansion policies: Evidence from Help to Buy". In: *Journal of Urban Economics* 139, p. 103611. URL: https://doi.org/10.1016/j.jue.2023.103611.

Cattaneo, Matias D., Michael Jansson, and Xinwei Ma (2018). "Manipulation Testing Based on Density Discontinuity". In: *The Stata Journal* 18.1, pp. 234–261. URL: https://doi.org/10.1177/1536867x1801800115.

Cengiz, Doruk et al. (2019). "The Effect of Minimum Wages on Low-Wage Jobs". In: *The Quarterly Journal of Economics* 134.3, pp. 1405–1454. URL: https://doi.org/10.1093/qje/qjz014.

Central Bank of Ireland (2016). *Review of Residential Mortgage Lending Requirements*. Tech. rep. Central Bank of Ireland. URL: https://www.centralbank.ie/docs/default-source/financial-system/financial-stability/macroprudential-policy/policy-documents/2016-review-of-residential-mortgage-lending-requirements.pdf.

Central Statistics Office (2026). *Market-based Household Purchases of Residential Dwellings*. URL: https://data.cso.ie/table/HPM03.

Citizens Information (2025). *Stamp Duty on Property*. URL: https://www.citizensinformation.ie/en/housing/owning-a-home/buying-a-home/stamp-duty/.

Clemente-Casinhas, Luís and Sofia Vale (2026). "Exemption or Illusion? The Impact of a Youth Tax Policy on House Asking Prices in Portugal". In: *Real Estate Economics* 54.2, pp. 469–503. URL: https://doi.org/10.1111/1540-6229.70020.

Competition and Consumer Protection Commission (2025). *Room for Improvement: Examining Information Availability in Residential Property Purchases*. URL: https://www.ccpc.ie/business/wp-content/uploads/sites/3/2025/07/CCPC-Homebuying-Research-Report.pdf.

Department for Levelling Up, Housing and Communities (2024). *Help to Buy: Equity Loan Scheme: Data to 31 May 2023*. URL: https://www.gov.uk/government/statistics/help-to-buy-equity-loan-scheme-data-to-31-may-2023/help-to-buy-equity-loan-scheme-data-to-31-may-2023.

Guren, Adam M. (2018). "House Price Momentum and Strategic Complementarity". In: *Journal of Political Economy* 126.3, pp. 1172–1218. URL: https://doi.org/10.1086/697207.

Hargaden, Enda Patrick (2017). *The Direct and Spillover Effects of Taxation: Evidence from a Property Tax Break for First-Time Buyers*. URL: https://www.aeaweb.org/conference/2018/preliminary/paper/r347bh9k&ved=2ahUKEwj3opPP2IqTAxXGBfsDHaqeAQEQFnoECB8QAQ&usg=AOvVaw3uWOlg_8OOZbnDzakSubgq.

Kleven, Henrik J. (2016). "Bunching". In: *Annual Review of Economics* 8.1, pp. 435–464. URL: https://doi.org/10.1146/annurev-economics-080315-015234.

Kleven, Henrik J. and Mazhar Waseem (2013). "Using Notches to Uncover Optimization Frictions and Structural Elasticities: Theory and Evidence from Pakistan". In: *The Quarterly Journal of Economics* 128.2, pp. 669–723. URL: https://doi.org/10.1093/qje/qjt004.

Kopczuk, Wojciech and David Munroe (2015). "Mansion Tax: The Effect of Transfer Taxes on the Residential Real Estate Market". In: *American Economic Journal: Economic Policy* 7.2, pp. 214–257. URL: https://doi.org/10.1257/pol.20130361.

Krolage, Carla (2023). "The Effect of Real Estate Purchase Subsidies on Property Prices". In: *International Tax and Public Finance* 30. URL: https://doi.org/10.1007/s10797-022-09726-0.

Lambie-Hanson, Lauren, Wenli Li, and Michael Slonkosky (2022). "Real Estate Investors and the U.S. Housing Recovery". In: *Real Estate Economics* 50.6, pp. 1425–1461. URL: https://doi.org/10.1111/1540-6229.12396.

Leung, Charles Ka Yui, Tin Cheuk Leung, and Kwok Ping Tsang (2015). "Tax-Driven Bunching of Housing Market Transactions: The Case of Hong Kong". In: *International Real Estate Review* 18.4. URL: https://doi.org/10.53383/100210.

McCann, Fergal and Anuj Pratap Singh (2023). "Housing Assistance Policy for Mortgage Borrowers: Liquidity Improvements or Price Acceleration?" In: *SSRN Electronic Journal*. URL: https://doi.org/10.2139/ssrn.6293140.

Polyakova, Maria and Stephen Ryan (2019). "Subsidy Targeting with Market Power". In: *NBER Working Paper* 26367. URL: http://www.nber.org/papers/w26367.

Property Services Regulatory Authority (2026). *Residential Property Price Register*. URL: https://www.propertypriceregister.ie/.

Revenue Commissioners (2024). *Help to Buy (HTB)*. URL: https://www.revenue.ie/en/tax-professionals/tdm-wm/income-tax-capital-gains-tax-corporation-tax/part-15/15-01-46.pdf.

– (2026). *Annual cumulative claims by build, value, loan-to-value and amount claimed by year*. URL: https://www.revenue.ie/en/corporate/documents/statistics/tax-expenditures/annual-claims-by-build-value-ltv-and-amount-claimed-by-year.csv.

Sallee, James M. and Joel Slemrod (2012). "Car Notches: Strategic Automaker Responses to Fuel Economy Policy". In: *Journal of Public Economics* 96.11-12, pp. 981–999. URL: https://doi.org/10.1016/j.jpubeco.2012.06.005.

Slemrod, Joel, Caroline Weber, and Hui Shan (2017). "The Behavioral Response to Housing Transfer taxes: Evidence from a Notched Change in D.C. Policy". In: *Journal of Urban Economics* 100, pp. 137–153. URL: https://doi.org/10.1016/j.jue.2017.05.005.

Uribe, Juan Pablo (2022). "Equilibrium Effects of Housing Subsidies: Evidence from a Policy Notch in Colombia". In: *Job Market Paper*. URL: https://juanpablo-uribe.github.io/Uribe_JMP_Housing_Subsidies.pdf.

Welsh Government (2026). *Help to Buy - Wales (Shared Equity Loan Scheme): October to December 2025*. URL: https://www.gov.wales/help-buy-wales-shared-equity-loan-scheme-october-december-2025.